

Organizational Technology Ladders: Remote Work and Generative AI Adoption

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Abstract

In this study, I propose that firms move along an “organizational technology ladder”: adopting one technology transforms hiring and work processes, and builds skills and capital which then change the cost of adopting the next technology. I focus on the impact of firms’ adoption of remote work technology on the proliferation of generative AI technology: using detailed job posting data and an IV approach based on differences in labor market pressure to allow remote work, I show that a 10 pp rise in remote work induced by the instrument in 2021–22 causes an increase in job postings mentioning generative AI skill by 2024 of 0.4 pp across firms and 1.1 pp across occupations within firms. Generative AI adoption also changes firms’ use of remote work: a synthetic difference-in-differences estimation around the release of ChatGPT shows that firms with greater exposure to generative AI reduced their remote hiring by 13% relative to a matched control group, with a more negative effect for tech firms than for non-tech firms. I explore the mechanism for these effects and show that remote work adoption shifts hiring towards technology and management skills that then enable more rapid generative AI adoption. Generative AI adoption skews remaining human tasks towards requiring more decision-making in the tech sector, which can explain the decline in remote work in tech, if decision-intensive tasks are less productive while working remotely. Firms that have low remote productivity, as evidenced by having a return-to-office mandate,

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are more likely to adopt generative AI after hiring remotely. I formalize this technology ladder mechanism using a task-based model of sequential firm investments in new technologies that change the importance of in-person coordination.

In November 2022, just as firms were settling into post-pandemic work arrangements, ChatGPT’s release triggered one of the fastest waves of technology adoption in history: Within months, generative AI tools spread through organizations at unprecedented speed (Bick, Blandin, and Deming, 2024), spurred by the potential to automate a large share of tasks (Eloundou, Manning, Mishkin, and Rock, 2023) and by field experiments documenting large productivity gains (Noy and Zhang, 2023; Brynjolfsson, Li, and Raymond, 2023). Yet adoption has been strikingly uneven. Some firms rapidly integrated these tools while others lag behind, creating early productivity gaps that may lead to persistent advantages Bloom, Hassan, Kalyani, Lerner, and Tahoun (2021). Understanding why requires looking beyond generative AI in isolation to examine how firms’ prior technological choices shaped their readiness for this technological transformation.

This paper uncovers a fundamental dynamic in how organizations navigate technological change: firms move along an “organizational technology ladder” where each adopted innovation transforms their capacity to embrace the next. The massive shift to remote work during COVID-19 didn’t just change where people worked—it also fundamentally altered firms’ technological capabilities, hiring patterns, and organizational structures in ways that determined their ability to embrace generative AI when it arrived. Organizations that climbed the earlier rungs of the ladder, for example, by successfully adapting to remote work, positioned themselves to ascend to the next rung when generative AI opportunities became salient.

My findings show a strong pattern of technological interdependence: Firms forced to accommodate more remote work during the pandemic subsequently adopt generative AI at significantly higher rates. I show that a plausibly exogenous 10 percentage point increase in remote work causes a 0.4-1.1 percentage point increase in generative AI adoption. Moreover, some firms, particularly in the tech sector, then reduce their remote hiring after implementing AI tools: I show that high exposure to generative AI on average leads to reduction in remote hiring after ChatGPT’s release.

I argue that these empirical patterns can be explained by the organizational technology ladder operating through three key mechanisms, which I formalize in a task-based model of technology adoption. First, infrastructure complementarity, where remote work forces investments in digital systems that generative AI later builds upon; Second, skill accumulation: adapting to remote work requires hiring technical and managerial talent that then enables AI implementation. Third, workflow transformation: distributed work creates processes that facilitate AI integration, but generative AI also shifts the remaining human tasks toward complex decision-making tasks that benefit from in-person collaboration.

My empirical strategy exploits exogenous variation in *how much* remote work different

firms were forced to accommodate during the pandemic. Specifically, I exploit exogenous variation in firms' exposure to labor market pressure to provide remote work options as a benefit to workers: The main instrument in my IV estimations is based on the firm's exposure to labor markets with competition from *other firms* that are more likely be offering remote work as a perk for their workers. An alternative instrument uses a firm's pre-pandemic exposure to labor markets with long commuting times as a proxy for greater employee demand for remote options during the pandemic. I also identify exogenous within-firm variation in remote work across different occupations by further interacting these firm-level instruments with national occupational trends in remote work.

Using these instruments, I provide causal estimates of the effects of a higher rate of remote work at a firm on its adoption of generative AI. I provide descriptive evidence that greater remote work is associated with higher generative AI adoption, but that remote work shares declined after generative AI became salient. Using the IV approach, I find that, across firms, a 10 pp higher remote work share causes firms to increase their hiring for generative AI skills by at least 0.4 pp by 2024. Within firms, a 10 pp difference in remote work across occupations is associated with a 1.1 pp increase in generative AI mentions when hiring. These effects are robust to controlling for a flexible set of control variables that capture firms' fundamental suitability for remote work and generative AI automation, their technology capabilities before the pandemic, as well as industry sector fixed effects; and also to using the alternative IV strategy based on variation in commuting time. Moreover, within-firm estimates control for both firm and occupation fixed effects, and results are robust to geographically disaggregating the data and including MSA fixed effects.

However, technology interactions do not just operate in one direction, as the incentives to continue using earlier technologies changes with the adoption of new ones. To show this, I estimate the effect of exposure to generative AI on remote hiring after the release of ChatGPT. I use a synthetic difference-in-differences design following Arkhangelsky, Athey, Hirshberg, Imbens, and Wager (2021), which compares the hiring behavior of firms with high (top decile) generative AI automation potential to firms with comparable remote work trends before the release. I find that high generative AI exposure firms significantly reduce their remote hiring share after the ChatGPT release relative to the control group. Greater exposure to generative AI led to a reduction in remote hiring by about 13% for the most exposed firms relative to a matched synthetic control group. This effect results from a reduction in the level of hiring across all positions, with a disproportionate decline of remote jobs. However, the negative effect is predominantly driven by declines among firms in the tech sector, with non-tech firms having either small positive or zero effects of generative AI use on remote work.

Third, I develop a simple conceptual framework that captures the potential channels for *why* remote work and generative AI technology adoption are connected and generates testable predictions for *how* firms change their organizations when they adopt technological innovations. This model builds on the task-based approach to modeling automation (Acemoglu and Restrepo, 2022; Thompson and Autor, 2024) but incorporates the organizational effects of the two technologies of interest: I model the effect of remote work as an increase in productivity by the worker that comes at the cost of lower decision-making quality. Generative AI is modeled as an automation technology that reduces the tasks done by humans, and which leaves tasks to the human that are more or less sensitive to good decision-making. This “judgment skew” of automation can explain the existence of a *technology substitution effect* and the heterogeneity between tech and non-tech sectors: if generative AI raises the average decision intensity of the remaining human tasks, this makes it more costly to continue operating remotely if decision-making quality is lower while remote.

In this framework, the cost of adopting each technology depends on the existing levels of information technology at the firm. Thus, if remote work leads firms to invest in technology skill, it may thereby lower the incremental investment needed to implement generative AI. Intuitively, as long as these complementarities in technical infrastructure are large enough, a *technology ladder* exists: remote work adoption makes generative AI easier to implement.

Fourth, I show empirically that adaptation to remote work matters for whether firms invest in generative AI: The mechanism for the technology ladder is supported by findings that remote work adoption during the pandemic causes firms to increase hiring for roles that are associated with higher decision-making skills, social skills and managerial responsibility; and also increases investments in technology skills—and these skills are associated with a greater adoption of generative AI for a given exposure to the technology’s benefits. I show that, in general, technology ladder effects are stronger for firms that previously hired for more technology skills, for more complex roles and for more managerial roles. As a case study for the importance of adaptation, I show that firms that issued return-to-office mandates—as a proxy for perceiving remote work to have relatively low productivity—have a significantly higher likelihood of investing in generative AI skills in response to higher remote work shares.

Conversely, generative AI exposure is less likely to result in a decline in remote hiring for firms with more managerial capacity, roles that require more social interaction, and with jobs that are less decision-intensive. These findings align with the idea that organizations that can better overcome the communication barriers associated with remote work are then less likely to embrace generative AI tools—compared to firms that struggle with remote processes.

Fifth, I show that the difference between the generative AI effects on remote work in the tech and non-tech sector (very negative in tech, and small in other sectors) can be explained by a greater increase in demand for decision-making skills in the tech sector after generative AI is adopted. This aligns with the proposed conceptual framework, in which automation with a low judgment skew, such that human decision-making intensity increases in the remaining human tasks, results in remote work becoming less tenable.

These results confirm the existence of an “organizational technology ladder”: firms that adopt one technology—remote work in this case—then benefit more from subsequent technology waves—here, generative AI—that build on the same investments in digital infrastructure and require similar changes in work processes. This path dependency can further exacerbate differences in productivity among firms as the effects of the two biggest technological changes in firms’ work processes during the last decade amplify one another in some organizations, and are dampened in others.

Related Literature. The proposed notion of a “technology ladder” that operates through organizational adaptation builds on earlier studies which showed that technology can affect the organization of firms and thereby change wage inequality (Garicano and Rossi-Hansberg, 2006), and that technologies differ in their effect on the centralization of decision-making within firms (Bloom, Garicano, Sadun, and Van Reenen, 2014). Moreover, there is existing evidence that the effect of technologies on productivity depends on the degree to which complementary organizational changes are made (Bresnahan, Greenstein, Brownstone, and Flamm, 1996; Brynjolfsson and Hitt, 2000). For example, Brynjolfsson and McElheran (2016) show that greater IT investments and more educated workers are associated with greater adoption of data-driven decision-making by firms in the 2010s. Similarly, Brynjolfsson, Rock, and Syverson (2021) argue that complementary firm investments in reorganization and implementing new decision processes are critical for unlocking the productivity benefits of new technologies. Bresnahan, Brynjolfsson, and Hitt (2002) argue that this organizational adaptation can reinforce the effect of technologies on labor demand. The contribution in this paper is to propose a link between the effects of adopting one technology and the likelihood and effectiveness of adopting another, and also to provide detailed empirical evidence on the firm investments in skills that accompany technology adoption.

While this paper is, to my knowledge, the first study to draw a direct link between remote work and generative AI adoption, previous studies have found that remote work benefits from existing investments in communication technology (Boeri, Crescenzi, and Rigo, 2024) in line with evidence that remote work creates more communication overhead (Bao, Li, Xia, Zhu, Li, and Yang, 2022; Gibbs, Mengel, and Siemroth, 2023). My finding that greater remote work may be associated with greater automation is also consistent with a literature

that finds that remote work changes the attachment between firms and their employees, leading to a decline in the quality of employee engagement with co-workers (Gibbs et al., 2023; Akan, Barrero, Bloom, Bowen, Buckman, Davis, Pardue, and Wilkie, 2024; Emanuel and Harrington, 2024).

My findings regarding the effect of generative AI on hiring relate to an emerging literature that studies the effect of generative AI adoption on firms: Gulati, Marchetti, Puranam, and Sevchenko (2025) show that generative AI adoption in particular roles within firms is associated with higher demand for cognitive and social skills in job postings for those roles. Alekseeva, Azar, Giné, and Samila (2024) show that non-generative AI adoption was also associated with greater hiring for managerial roles and demand for managers with more interpersonal and cognitive skills. Humlum and Vestergaard (2025a) show, using survey data, that the introduction of generative AI changes the task structure of the affected jobs.

This paper has the following structure: the next section discusses the data that I use. Section II presents the empirical approach to estimating the causal effects of remote work adoption on generative AI and vice versa. Then, I provide a set of new stylized facts regarding the relation between remote work and generative AI in Section III.A; estimate the causal effect of remote work on generative AI adoption in Section III.B, and the effect of generative AI exposure on remote work in Section III.D. To interpret the empirical findings, Section IV provides a simple conceptual framework, and in Section V I provide additional evidence on the heterogeneity of the effects and the changes in hiring that result from remote work and generative AI adoption. Finally, Section VI draws out the implications of these results and concludes.

I. Data

A. *Key variables*

Both hiring activity and investments in Generative AI skills and other characteristics of new jobs are measured using job postings data from Lightcast.¹ This data consists of the near-universe of online job postings in the U.S. from Jan. 2010 to Sep. 2024. I filter job postings in the following way: (1) Drop job postings by staffing companies. (2) Drop all jobs flagged as internships. (2) Retain only full-time jobs, dropping part-time jobs. I crosswalk all occupational data to SOC 2010 codes and refer to these as “occupations” throughout.

Remote work share. The job posting data allows for the identification of both jobs that are fully remote and jobs that involve a hybrid format with some presence in an office.

¹The Lightcast data vintage is as of October 18th, 2024.

The classification of *remote* jobs used in most of the analyses below represents all jobs that indicate either of these possibilities. For robustness checks, I also construct versions of the remote job definition where only *hybrid* or only *fully remote* jobs are included. When job postings are aggregated into firm or occupation or firm-by-occupation level panels, the counts reflect the number of job postings *posted* in each category in the time period, which does not necessarily reflect actual hiring, but rather proxies for hiring demand.

Generative AI adoption. While I cannot observe the *usage* of Generative AI within firms, the job posting data allows me to see when firms are either requiring Generative AI-related skills from new workers, or describing the work activities in a job as involving Generative AI tools. Therefore, I will use mentions of Generative AI and related technologies in job postings as a proxy for the degree to which those jobs involve using Generative AI-related tools. Labels for "Generative AI" skills mentioned in a job posting are supplied by Lightcast based on the job posting text.

Generative AI exposure. As a proxy for generative AI exposure based on the composition of the tasks done in a firm, I use the Eisfeldt, Schubert, Zhang, and Taska (2023) measure of occupation exposure to generative AI. This measure uses the approach in Eloundou et al. (2023) to first evaluate the task-level ability to potentially deploy generative AI productively and then aggregates from the set of tasks in an occupation to an occupation-level exposure score that represents the weighted share of tasks that could be made more productive with generative AI. I compute the exposure of a firm’s hiring in a given year as a job posting weighted average over the exposure of the occupations that it is hiring for.

B. Other data

Occupational wage and employment data. I obtain data on employment and average wages by occupation and MSA from the Bureau of Labor Statistics’ Occupational Employment Statistics. I crosswalk all occupational data to SOC 2010 codes and Core-Based Statistical Area codes (CBSAs).

Occupation characteristics. To evaluate whether the estimated effects differ between occupations with different characteristics, I use O*Net data on the characteristics of different occupations to construct summary measures that assess the prevalence of different skills in different occupations. I construct the following measures of occupation characteristics: To capture the importance of “decision-making” in a job, I average the level of related work activities involved in an occupation according to O*Net, based on Deming (2021).² An indicator of social skills required for a job are constructed as described in Deming (2017). I

²These activities are: Making decisions and solving problems; Developing objectives and strategies; Organizing, Planning and prioritizing work.

also construct a measure of job inflexibility based on Goldin (2014), and a measure of leadership skills based on Schubert, Stansbury, and Taska (2019), which is an average of the work activities for which a high value best predicts managerial leadership positions. I also use a proxy for the remote work suitability of an occupation—the “teleworkability” measure constructed by Dingel and Neiman (2020).

Skill demand. Firm- and occupation-by-firm measures of skill demand are constructed directly from the job posting data by using tags provided by Lightcast for job postings that require a college degree, an advanced degree, or mention different skills. I focus on communication, decision-making, data management, data science, machine learning, and deep learning skills. I also construct an indicators for jobs that require “high” experience when at least 5 years of experience are required.

Commuting cost. To measure differences in commuting cost across different MSAs, I aggregate county-level data (using population weights) on average travel time to work by county from IPUMS NHGIS (Manson, Schroeder, Van Riper, Knowles, Kugler, Roberts, and Ruggles, 2024), which is based on the 2015-2019 American Community Survey data.

Return-to-Office policies. I obtain information on firms’ return-to-office (RTO) policies by collecting crowdsourced public information from the website Flex Index. I match companies listed on Flex Index to companies in Compustat using company names and locations. The final dataset includes data on 1,336 companies and their office presence requirements. I classify a company as having provided a “Return-to-Office” mandate, if its RTO policy requires either a full-time presence, or specifies a non-zero minimum number of days in the office.³

II. Empirical Approach

To test whether there is in fact a “technology ladder effect”, where one technology’s adoption impacts the use of the other by firms, we want to estimate the effect β of having a greater share of remote jobs on firms’ investment in generative AI skills. This effect is estimated from regressions of the form

$$\text{GenAIJobShare}_i = \alpha + \beta \text{RemoteWorkShare}_i + \text{Controls}_i + \varepsilon_i, \quad (1)$$

where the unit of observation i can be a firm or an occupation-by-firm unit. The Controls_i are detailed below when discussing how I address identification concerns. While the main

³See Appendix B for more information on the Flex Index data cleaning and the method for matching to Compustat.

estimation will be cross-sectional, using job posting data for the 12 months ending in, and including, September 2024, to construct the dependent variable, some of the control variables are constructed in earlier periods to capture past characteristics of the firm, occupation, or location.

A. *Remote work and Generative AI: identification issues*

Why would having many workers in remote positions *cause* a firm to adopt generative AI at a faster rate? To structure the empirical approach, it is important to first clarify the potential causal channels of interest and empirical challenges in identifying them.

The effects of remote work on generative AI adoption are potentially ambiguous: generative AI tools could complement remote workers more than in-office workers, for example building on their existing familiarity with digital workflows, or enabling better remote work tools. However, if generative AI is more likely to automate remote work tasks entirely, the use of these tools can also substitute for remote workers. It is therefore an empirical question whether a greater use of remote workers is likely to lead to more or less investment in generative AI skills by firms. The organizational technology ladder mechanism where remote work adoption *causally* affects generative AI adoption can arise from a number of different channels:

1. *Augmentation*: Remote work jobs might benefit directly from applying generative AI tools, and this greater benefit of adoption drives higher demand for generative AI skills.
2. *Digital infrastructure*: the digital infrastructure, data processing, and other technical capacity improvements that result from implementing a remote work-friendly organization, make it easier to adopt generative AI technologies.
3. *Workflow restructuring*: by making workflows modular, asynchronous, decentralized, and not reliant on real-time in-person input, while incorporating mechanisms for remote quality validation, remote-adapted firms are more suitable for inserting automated AI tools into the value chain.
4. *Automation*: For some firms, remote workflows are less suitable, but they find themselves unable to enforce a post-pandemic return to in-person work. In that case, automation through generative AI can provide a channel for organizational adjustment that allows firms to reduce the share of their work done remotely.

On the other hand, there are also possible mechanisms through which remote work and generative AI adoption may coincide in the same firms and occupations, but without a causal link from one to the other:

4. *Task characteristics*: it is possible that the tasks that are amenable to being done remotely also happen be the ones for which generative AI is useful. In that case, greater adoption of both technologies is driven by greater suitability for the technologies being correlated.
5. *Innovation orientation / tech-savviness*: some firms may have organizational cultures and leadership that are more open to, and capable at, embracing new technologies and being at the leading edge of innovation. This trait may make these firm more likely to experiment with, and adopt, both remote work and generative AI as the newest waves of technological progress.

One key empirical issue is trying to distinguish these causal and non-causal channels. To see that this is a relevant concern, I compare a proxy for the remote work suitability of an occupation—the “teleworkability” measure by Dingel and Neiman (2020)—to the measure of generative AI exposure at the occupation level from Eisfeldt et al. (2023). Figure A.4, Panel A, shows that there is a positive relationship between occupations having tasks that are suitable for remote work and tasks that are exposed to generative AI capabilities.

The ideal setting for estimating the effect of remote work adoption on generative AI adoption would therefore require comparing groups of jobs that have similar *suitability* for both remote work and generative AI deployment, and also are at companies with similar “tech-savviness” or innovative capacity. The ideal experiment to identify causal effects would then require one group of these jobs to experience greater prevalence of remote work for an exogenous reason, so that we can compare generative AI adoption rates to see what the magnitude of the causal channel operating *through* remote work adoption is.

I approximate this natural experiment by controlling for potential confounders and instrumenting for remote job prevalence using exogenous variation that is plausibly unrelated to firm-level differences in unobserved characteristics that might be driving generative AI adoption. As control variables, I include Dingel and Neiman (2020) teleworkability and Eisfeldt et al. (2023) generative AI exposure scores to capture the intrinsic nature of the tasks performed in different firms and occupations that might make them more or less suitable for adopting these technologies, as well as controls for the average education level of a firm’s hiring and for the firm’s hiring for specific technology skills that might capture the ability of employees to implement innovations.⁴ Moreover, the most stringent specifications include

⁴See Section III.B for the full list of baseline controls.

fixed effects at the level of the firm and for each occupation that capture general differences in tech-savvy, or in the tendency to adopt either of the technologies of interest.

B. Remote work instruments

To identify exogenous variation in the remote job share at the firm level, I exploit a novel source of variation in whether a firm adopts remote work at a high rate in a particular location: the interaction between *labor market pressure* and the *ability* to let employees work remotely. That is, many studies (e.g. Maestas, Mullen, Powell, Von Wachter, and Wenger (2023), Powell and Wenger (2024)) have found that workers consider the ability to work remotely as a sizable non-monetary benefit. Other studies have found that offering hybrid work options increases worker retention (Bloom, Han, and Liang, 2024), and that remote workers are more productive as a result of shorter commute times and better work-life balance (Choudhury, Foroughi, and Larson, 2021).

As a result, similar to the way that workers' outside options encourage firms to match wages on offer elsewhere (Schubert, Stansbury, and Taska, 2024), labor market competition induces firms to offer remote work options. This pressure should be larger in labor markets where *other* employers are likely offering this perk, or where commuting is more burdensome and workers therefore likely perceive a larger benefit of working from home (Flynn, Ghent, and Nair, 2024). Based on this intuition, I construct two novel instruments for remote work.

Main instrument. While the actual adoption of remote work by other employers in particular labor markets is both difficult to observe (as remote job locations are, by definition, not well defined in job postings), and might be simultaneously determined with post-pandemic choices by the firm of interest, the *average pre-pandemic ability* to work remotely in the labor markets that a firm hires in is both observable and unlikely to be driven by the focal firm's ex post adoption behavior. Thus, I construct a firm-level instrument for remote work adoption based on labor market competition as

$$\begin{aligned} Z_f^T &= \left(\sum_m \phi_{fm,2019} T_{m,2019} \right) \times T_{f,2019} \\ &= \underbrace{\text{Avg. Labor Market Teleworkability}_{f,2019}}_{\text{Remote work adoption potential of the firm's labor market pre-Covid}} \times \underbrace{\text{Firm Teleworkability}_{f,2019}}_{\text{Firm remote work adoption potential pre-Covid}}, \end{aligned}$$

where $T_{m,2019}$ is the average Dingel and Neiman (2020) teleworkability among job postings in a particular MSA based on 2019 job postings, and the MSAs are weighted by the share $\phi_{fm,2019}$ of all firm f hiring done in each location m as of 2019. $T_{f,2019}$ is the average

teleworkability among job postings by firm f in 2019.

Alternative instrument. To proxy for the pressure to permit remote work during and after the pandemic that arises from workers' perceived cost of commuting, I use estimates of the pre-pandemic average travel time to work in each MSA, based on ACS data from 2015-2019 (similar to the commuting distance instrument for remote work used by ?). I aggregate these geographic proxies to a firm level measure of commuting cost using the firm's share of hiring in different locations in 2019. As the desire by employees to work remotely will only translate into remote jobs if positions at a firm are teleworkable, I again interact this measure of remote work pressure with the firm's teleworkability when constructing the commuting cost instrument:

$$\begin{aligned}
Z_f^C &= \left(\sum_m \phi_{fm,2019} C_{m, '15-'19} \right) \times T_{f,2019} \\
&= \underbrace{\text{Avg. Labor Market Commuting Time}_{f,2019}}_{\substack{\text{Travel time to work} \\ \text{in the firm's labor market pre-Covid}}} \times \underbrace{\text{Firm Teleworkability}_{f,2019}}_{\substack{\text{Firm remote work adoption} \\ \text{potential pre-Covid}}}
\end{aligned}$$

where $C_{m, '15-'19}$ is the average travel time by MSA, and the MSAs are weighted by the share $\phi_{fm,2019}$ of all firm f hiring done in each location m as of 2019.

Occupation-by-firm instruments. As I also want to be able to estimate exogenous variation across occupations *within firms*, I construct occupation-by-firm-level instruments based on a similar intuition. For occupations where the hybrid work share was higher at the national level post-Covid (but before the ChatGPT release) in 2021-2022, employers who are able to let their workers work remotely likely face more labor market pressure to grant this perceived perk. Thus, adoption of remote work in an occupation-by-firm cell can vary in response to this labor market channel, independent of overall firm incentives to adopt generative AI. To be specific, I construct occupation-by-firm instruments by interacting the firm-level instruments with the hybrid work share in the occupation during the late pandemic years 2021-2022:

$$\begin{aligned}
Z_{fo}^T &= Z_f^T \times \text{HybridWorkShare}_{o, '21-'22} \\
Z_{fo}^C &= Z_f^C \times \text{HybridWorkShare}_{o, '21-'22}
\end{aligned}$$

Here, the hybrid work share is used (rather than the share of all remote jobs) as this seems to have a better predictive power than the total remote job share, perhaps because fully remote job hiring is less driven by local labor market competition (as it is, by definition, not limited to hiring in specific markets). Moreover, I construct the national occupation hybrid work shares as a leave-one-out measure that omits each firm's own hiring in that

occupation, in order to avoid mechanical correlation between the national trend used in the instrument and remote hiring for large employers (Borusyak, Hull, and Jaravel, 2022).

Exclusion restriction. These shift-share instruments rely only on pre-Covid job compositions in different locations, firms and occupations, and exogenous national trends in remote work before generative AI technology was salient and are therefore unlikely to be correlated with endogenous generative AI adoption choices by a firm, other than through the remote work channel. Formally, the relevance condition in this setting requires that the *interaction* between labor market suitability for remote work (due to competition or commuting costs) and a firm’s ability to let workers go remote drives remote work adoption in a way that is not explained by either overall labor market suitability for remote work, or firm suitability on their own—as the latter uninteracted variables are included as control variables in the estimation. The exclusion restriction then requires, for example, that if we compare two firms that have similar teleworkability, similar remote shares pre-pandemic, similar exposure to generative AI, are in the same industry, and have similar observable hiring for technology skills, then the one that finds itself in a labor market with more other firms that have high teleworkability does not systematically differ from the other firm in a way that drives generative AI adoption *except* through greater remote work adoption during the pandemic.

At the occupation-by-firm level, the exclusion restriction requires that greater pressure to adopt remote work, arising from high national hybrid work shares in an occupation, leads to differential generative AI adoption for some occupations, but only within firms that face higher overall labor market pressure to allow remote work, and that this effect operates only through a remote work channel. Moreover, in this setting we are able to condition on a firm’s overall tendency to adopt generative AI, as well as an occupation’s overall tendency to use generative AI, by including the corresponding fixed effects in the estimation.

As a firm’s teleworkability or exposure to particular labor markets on their own might be correlated with particular unobservable firm characteristics, the IV estimations will always control for the uninteracted components of the instrument.

Unit of observation. Depending on the particular dimension of the variation in generative AI adoption that we are interested in, a different unit of observation is appropriate. One question is whether the *same firms* are seeing higher remote work adoption and generative AI adoption, which matters for inequality in productivity across the economy as some firms might see compounding effects of technology, and might also provide evidence about the degree to which there is variation in corporate strategy across firms. This question is best studied using firm-level evidence. At the same time, there are questions of whether generative AI adoption is more prevalent in the same occupations where remote work was

adopted, which speaks to the degree to which these technologies complement or substitute for one another *within* particular jobs, holding a firm’s overall hiring and work processes constant. I will use a firm-by-occupation sample to study the latter questions, such that I can identify differences in effects across occupations within the same firm. In a robustness check, I also analyze data at the firm-by-MSA and at the firm-by-occupation-by-MSA level based on the “location” of a job posting—however, the location indicator for remote work is, by definition, not necessarily reliable or binding, and the IV approach is therefore not as compelling at this level of granularity, which is why these results are not one of the main specifications.

C. Estimating the effect of generative AI adoption on remote hiring

The stylized fact pattern shown in Panel B of Figure 1, and described in Section III.A, shows that higher later generative AI adoption is associated with a strong reversal in remote work shares around the time when ChatGPT was released. I want to test more formally whether these differential trend breaks in remote work shares around the ChatGPT release period might be driven at least in part by generative AI innovations.

Identifying whether this association in trends is likely to represent a causal “technology substitution effect” requires an empirical design that can address the identification issues mentioned before. Given the surprising release of ChatGPT in November 2022 and rapid rise in adoption afterwards (see Figure 1), an event study methodology, such as differences-in-differences might be appropriate in this case and has been used to study the effects of generative AI in other contexts (e.g Eisefeldt et al. (2023)). Conceptually, this corresponds to determining the treatment effect τ of being in a high generative AI exposure group in a regression equation of the form

$$\text{RemoteWorkShare}_{it} = \mu + \alpha_i + \gamma_t + \tau \mathbb{1}[\text{Post-ChatGPT}]_t \times \mathbb{1}[\text{High GenAI Exposure}]_i + \varepsilon_{it}, \quad (2)$$

which includes firm fixed effects α_i and time fixed effects γ_t , which absorb level differences between firms and national trends in the remote work share. Here, I am using the exposure of a firm to generative AI benefits as the treatment variable, because the actual adoption of generative AI in response to exposure will be an endogenous response that is in part driven by a firm’s adaptation to remote work—which is one of the core arguments of this paper. The exclusion restriction is therefore more plausible when estimating the intention-to-treat effect of pre-determined exposure to generative AI around the plausibly exogenous ChatGPT release date.

However, estimating a causal effect using a differences-in-differences approach normally

requires a parallel trend assumption to hold (see, e.g. Rambachan and Roth (2023)), which assumes that treatment and control group can be expected to behave similarly in the counterfactual scenario where no treatment occurs. As the descriptive analysis shows, generative AI adoption is clearly not randomly assigned with regard to remote work shares—in fact, I argue in this paper that higher remote work adoption eventually *causes* greater generative AI adoption. Moreover, there is a positive occupation-level correlation in pre-existing exposure to the two technologies—as shown in Appendix Figure A.4—which means that adoption will also be correlated. As a result, firms that have higher later generative AI adoptions likely also had steeper run-ups in remote work earlier in the pandemic. This difference in pre-trends can also be seen in Appendix Figure A.7, where I sort firms by their quartile of generative AI exposure. Comparing the firm-level remote work trends by generative AI exposure shows that firms with higher exposure have steeper run-ups in remote shares before the ChatGPT release. Any causal effect estimation of the effect going in the other direction—from generative AI adoption to remote work—therefore needs to try and hold remote work adoption trends constant when comparing firms around the time of the ChatGPT release event.

One could use control variables to try and capture the common variation in the confounding variable, which would mean trying to incorporate all the control variables that can capture potential drivers of differential remote work share trends at the firm level. However, as there is not necessarily a definitive list of these drivers, one would worry that the control and treatment group are not fully comparable in terms of their remote work trends even after including suitable observables.

Synthetic difference-in-differences. A more appropriate methodology in this case is therefore the synthetic difference-in-differences (SDID) design proposed by Arkhangelsky et al. (2021): instead of ex ante selecting which control variables might align remote work trends in the pre-event period between firms, in this methodology the control group is defined using optimal weights that are estimated in a first step designed to create a control group that matches the treatment group’s trend in the remote work share (as opposed to differences in levels which are absorbed into the firm-level fixed effects).⁵ A weighted average of this most-comparable group of firms is used to create a “synthetic” control group. The comparison of post-event relative trends between the treated and control groups then proceeds analogous to a difference-in-differences estimation, but weighting potential control group firms based on the optimal weights, and also applying time weights that optimally put higher weights on pre-periods that are more comparable to post-event periods for the

⁵I implement this estimation procedure using the `sdid` package in Stata, which is described in Clarke, Pailanir, Athey, and Imbens (2023).

control group.

Event study data. In order to use the SDID estimator, I need a balanced panel of company-by-period observations. I retain all firms that have at least 10 job postings in all quarters from Q1 2021 to Q3 2024, which is the sample period for the event study estimation, consisting of 7 quarters before and after the ChatGPT release in Q4 2022.

Treatment group. To define the treated group, note that *actual* adoption of generative AI is endogenous with regard to a firm’s characteristics. In contrast, a firm’s *exposure* to generative AI, based on its employment structure before ChatGPT was released, more plausibly represents an exogenous shock as it interacts with the exogenous timing of the release. I use the Eisfeldt et al. (2023) measure of occupation exposure to generative AI, and compute a firm’s 2022 exposure in its hiring as a job posting weighted average over the exposure of the occupations that it is hiring for. I sort firms by the continuous exposure variable and assign the unweighted top decile of firms by exposure to the “treated” group (and consider the effect of changing this cutoff threshold in robustness checks). Note that this means that the estimated effect does not compare firms with high exposure to firms with zero exposure, but rather to a synthetic control group that has *some* (albeit lower) exposure and very similar remote work pre-trends.

III. Remote Work and Generative AI Adoption

In this section, I first provide descriptive facts about the relation between remote work and generative AI that motivate the causal analyses. Second, I explore what firm characteristics are associated with greater adoption of remote work. Then, I provide evidence of the “technology ladder effect:” remote work adoption has a positive causal effect on the subsequent investment in generative AI skills. Finally, I estimate the effect of generative AI adoption on a firm’s hiring of remote workers—the “technology substitution effect”.

A. *Stylized facts: remote work and generative AI*

To understand whether firms are more or less likely to change their investments in hiring for generative AI skills if they already have a remote workforce, or might change their remote hiring with the emergence of generative AI tools, I start by providing descriptive evidence that these two technologies are likely to be connected. In this section, I collect a number of novel facts about remote work and generative AI skill demand:

1. **Industries with higher remote work prevalence also have higher generative AI adoption.** Appendix Figure A.1 shows the log of the remote share explains 43% of

the log generative AI adoption rate at the industry sector level

2. **Firms that hire more remotely also hire more for generative AI skills.** Figure A.2 shows in job postings data for the year-to-date (YTD) Sep. 2024 that the log of the remote share explains 13% of the cross-sectional variation in the log generative AI adoption rate across firms.
3. **Generative AI adoption rates are higher in remote jobs:** Appendix Table A.1 ranks 6-digit occupations by their prevalence of remote jobs (panel A) and generative AI (panel B) in job postings, as of YTD Sep. 2024, and shows the top 20 occupations. Occupations that adopt generative AI at high rates are similar between remote and non-remote workers, but generative AI adoption rates are higher among remote workers. For example, Technical Writer job postings mention generative AI in 6.1 percent of all job postings, but in 18 percent of remote job postings.
4. **Higher remote hiring during the pandemic predicts rapid generative AI adoption.** Appendix Figure A.3, panel A shows that occupations that adopted remote work at a faster rate from 2019 to 2022 also increased hiring more for generative AI-related skills from 2022-2024.
5. **Generative AI adoption coincides with post-pandemic declines in remote hiring.** Panel B of Appendix Figure A.3 shows that generative AI adoption from the ChatGPT release in Q4 2022 to Q3 2024 is negatively correlated with changes in remote work during the same time period among the largest occupations.
6. **Remote work adoption is higher for firms with technology skills.** Appendix Figure A.8 and Section C.A show that greater hiring for technology-related skills in 2019 is associated with greater adoption of remote work during the pandemic.

B. Remote Work Impact on Generative AI Adoption

To test the key hypothesis that there exists a “technology ladder effect,” this section estimates the causal effect of remote work on Generative AI adoption both across firms, and across occupations within firms.

First-stage. To visualize the relationship that underlies the variation in the instrument, Figure 2 plots the relationship between the competition IV instruments and the remote work shares at the firm and occupation-by-firm level, including a full set of control variables. As the figure shows, the instruments have a strongly positive relationship with the endogenous remote work prevalence.

Firm-level results. To what degree is the correlation between remote work shares at firms and their tendency to invest in hiring workers with generative AI skills causal? I use the IV approach described in Section II.B to estimate the equation

$$100 \times \text{GenAIJobShare}('23-'34)_i = \beta \text{RemoteWorkShare}('21-'22)_i + \alpha_{ind} + \text{Controls}_i + \varepsilon_i$$

for the generative AI share for the YTD Sep. 2024 period and the remote share during 2021-2022 (excl. Q4 2022). The results using the competition instrument are shown in Table II. Column (1) shows the firm-level OLS results, and column (2) the firm-level IV results. The coefficient corresponds to a % elasticity between the remote share and the generative AI skill share in the firm’s job postings, such that the IV specification in column (2) suggests that a 10 pp higher remote share causes about a 0.4 pp higher generative AI skill share. The baseline firm-level control variables include characteristics of the firm’s YTD Sep. 2024 hiring that consist of the average task-level teleworkability (Dingel and Neiman, 2020) and generative AI exposure (Eisfeldt et al., 2023), the level of education required for average job postings at the firm; and industry sector (NAICS 2-digit) fixed effects. They also include the firm’s pre-pandemic hiring characteristics, specifically the share of 2019 hiring that mentions each of a list of technology skills (data management, machine learning, deep learning, artificial intelligence) as a proxy for its tech-savviness, the firm’s 2019 remote hiring share and 2019 teleworkability and generative AI exposure. Moreover, baseline controls also include the characteristics of the firm’s hiring markets before the pandemic—which are constructed as firm hiring-weighted averages across MSA characteristics—specifically, exposure to MSA remote work shares and teleworkability. These baseline controls mean that the effect is unlikely to be driven by a sorting of jobs that are more suitable for both technologies into the labor markets that are affected by the exogenous remote work variation induced by labor market competition. Note that the IV estimates are substantially larger than the OLS estimates, suggesting that the bias removed by the IV approach is that the firms which endogenously choose to offer remote work tend to be less likely to adopt generative AI.

Within-firm results. Is the coincidence between remote work and generative AI driven only by firm-level dynamics, or is there also a causal effect of one on the other across occupations *within* firms? Table II, column (4) shows the results for this within-firm effect using the competition IV, additionally controlling for firm fixed effects and occupation fixed effects. This means that the effect is only identified off within-firm variation across occupations. The estimated effect suggests that higher remote work prevalence has a large positive causal effect across occupations within a firm (and across firms within an occupation), with a size in column (4) almost three times that of the across-firm effect. This provides additional confir-

mation that the across-firm results are not driven by some unobserved firm-level attitude towards innovation—as such an effect would be eliminated by the firm fixed effects included here.

These results also mean that, while differences in corporate processes and capabilities as a result of firm-level changes in response to remote work can perhaps explain part of the puzzle of why the adoption of these two technologies is correlated, this is not the full story. The inclusion of occupation and firm fixed effects means that the effects cannot be explained purely by a change in the composition of hiring by firms towards occupations that tend to use generative AI. The large within-firm causal effects suggest that *particular jobs* become more suitable for generative AI if those jobs themselves adopted higher rates of remote work at that firm.

C. Robustness checks: remote work effects on Gen. AI

This section explores whether the finding of a causal effect of remote work on generative AI adoption is robust to different empirical specifications.

Weak IV tests. Do we need to be worried about “weak instrument” bias in these estimations? Olea and Pflueger (2013) suggest constructing their “effective F” statistic, which is robust to heteroscedasticity, autocorrelation, and clustering, and to reject the hypothesis that weak instrument bias exceeds 10% of a worst-case benchmark at a 5% significance level when this effective-F statistic for the first stage exceeds an appropriate value. They show that this critical value is below 23.1, making 23.1 a conservative rule-of-thumb cutoff. In the setting that I study, with one endogenous regressor and one instrument, the effective F-statistic is identical to the Kleibergen Paap F-statistic (KPF), which I report in all IV regression tables. I find that the KPF is greater than 23.1 in all main analyses (see Table II), which means that it is unlikely that there is weak instrument bias.

Fully-remote and hybrid-only jobs. While the baseline estimation includes both fully remote and hybrid jobs, the Lightcast data allows me to confirm whether the causal relation between remote work and generative AI is robust to using alternative measures of remote work. Columns (1)-(4) of Table III show results analogous to the baseline IV estimates in Table II, but with remote work shares defined using only fully remote or only hybrid jobs.⁶ As the results show, the estimates are economically and statistically significant even when using these alternative definitions of “remote” work, suggesting that this finding is not sensitive to distinguishing precisely which form of remote work an employer offers.

⁶For the fully remote estimation in column (2), I use a variant of the instrument that replaces the leave-one-out hybrid job share in the occupation at a national level with the leave-one-out fully remote job share when constructing the firm-by-occupation level instrument.

Exclude tech sectors. One potential concern with the estimation is that there may be broader concurrent economic trends that happen to cause an increase in generative AI hiring precisely for those firms that also happened to be more inclined towards remote work for exogenous reasons. The baseline estimation already includes 2-digit industry fixed effects, eliminating the possibility that industry-level trends are driving the results. However, it could still be the case that the results are driven by the dynamics *within* only particular sectors. In particular, technology companies faced a dramatic rise and decline in hiring during and after the pandemic, which may lead their hiring responses to be distorted. Moreover, technology companies may be more responsive to a new technology trend—generative AI—than companies in other sectors, as they supply AI services themselves. To the degree that this differential response reflects organizational adaptation to technological innovation, it may also support the broad argument in this paper that pre-existing skill levels in a firm drive adoption. I use the definition from Acemoglu, Autor, Hazell, and Restrepo (2022) and label the information sector (NAICS sector 51) and the professional and business services sector (NAICS sector 54) jointly as the AI-related “tech sector.” In columns (5) and (6) of Table III, I allow this tech sector to have a different causal effect of remote work on generative AI adoption than other sectors. I find in column (5) that the effect of remote work on generative AI across firms is about a quarter larger within the tech sector than in other sectors. However, both within and across firms, the causal effect among non-tech sector firms is also economically and statistically significant—so the effect is by no means limited to tech firms. I further explore this heterogeneity by broad sector in Appendix Figure A.6, where I find that, when restricting the sample only to individual broad sectors, the estimated effects of remote work on generative AI adoption are largest in the tech sector (labeled “Professional & Technical Services”). However, they are consistently positive in all sectors, and are statistically significant at a 5% level in 5 out of 8 broad sectors.⁷

Alternative instrument. In the estimation in Table II, I used the instrument based on labor market competition from other teleworkable firms. However, there may be residual concerns that this instrument selects for firms that are unobservably different due to their sorting into labor markets with particular types of competitors. To allay this concern, I repeat the same estimation with the instrument based on average commuting time in a firm’s hiring labor markets. If one believes that this source of variation is more plausibly

⁷Firms are categorized as follows into broad sectors (using 2-digit NAICS codes): Professional & Technical Services (NAICS 51 and 54); Healthcare & Social Assistance (NAICS 62); Construction & Real Estate (NAICS 22, 23, and 53); Manufacturing (NAICS 21, 31, 32, and 33); Business Support & Financial Services (NAICS 52, 55, and 56); Education & Other Services (NAICS 61 and 81); Trade & Transportation (NAICS 42, 44, 45, 48, and 49); Hospitality & Entertainment (NAICS 71 and 72); Agriculture (NAICS 11). Note that Agriculture is omitted from the sectoral analysis due to insufficient sample size.

excluded from the estimation equation, conditional on the other controls, then the relevant IV estimates are shown in columns (7) and (8) of Table III: the effect estimates are also significant and positive and of a very similar magnitude to those in Table II. However, the first-stage Kleibergen-Paap F-statistics suggest that the commuting time IV is a weaker instrument than competition IV, which means that the latter yields more precise estimates. The key results in this paper are robust to using either of these instruments.

Geographic variation. An additional concern may be that the instrument based on labor market competition captures the association of particular geographic areas (e.g. the Bay Area / Silicon Valley) with both remote work and generative AI adoption. To allay this concern, I construct versions of my data that are disaggregated at the firm-by-MSA and firm-by-occupation-by-MSA level. Then, I repeat the baseline IV estimation, but including MSA fixed effects. To obtain an instrument that has MSA-level variation, I use 2019 MSA teleworkability interacted with firm-level teleworkability as the instrument in these specifications. Column (9) shows that the firm-level results are robust to including not just the baseline industry sector fixed effects but also MSA fixed effects. Column (10) shows that the estimated causal effect is also significant and large when including MSA and firm-level fixed effects such that effects use only within-firm variation across locations. Last, column (11) shows that (under the caveat that the instrument is somewhat weak in this dimension) there is even a significant effect of remote work on generative AI adoption across MSAs within a firm when we are able to control for occupation-level fixed effects.

D. Generative AI Effects on Remote Work

What happens to remote work after a firm adopts generative AI? As generative AI changes the task structure of jobs, we would expect firms to adapt by reorganizing their work processes. In this section, I provide firm-level evidence on how remote hiring at firms evolves in response to generative AI adoption.

The first question I explore is whether there is systematic evidence supporting the stylized fact pattern shown in Panel B of Figure 1, and described in Section III.A, of higher generative AI adoption being associated with a reversal in remote work use after ChatGPT was released. That is, I want to know whether this association reflects a causal “technology substitution effect.”

Generative AI exposure and generative AI adoption. To identify treated firms, I use generative AI exposure based on the task composition of jobs instead of observed generative AI adoption, in order to avoid the endogeneity issues associated with realized firm choices. However, this approach raises the question of whether the Eisefeldt et al. (2023)

measure of firm-level exposure to generative AI actually predicts firm-level adoption of generative AI. One way to validate this quasi-“first-stage” relationship is in the cross-section: Appendix Figure A.5 shows that there is a monotonically increasing relation between a firm’s predicted generative AI exposure and the actual share of generative AI skill mentions in all job postings. Moreover, the largest variation in adoption happens among the highest exposure deciles, which motivates focusing on the most exposed firms (rather than the least exposed firms) in the analysis.

Generative AI effects on remote work shares. How much does the remote job share change for high generative AI exposure firms after the ChatGPT release? Column (1) of Table IV shows the estimated post-ChatGPT effect of being in the top exposure decile on the remote share of job postings: the estimated effect on remote work shares is a reduction by 2.7 pp. This is a substantial effect: the treated group has a level of remote work of 20.3% in the pre-period (7.9% for the control group), so the post-release impact of being in the high exposure group corresponds to a 13% reduction on average in the remote share relative to the pre-period.

To verify that the synthetic difference-in-differences matching procedure actually results in a suitable control group, Figure 3, Panel (A) shows details on the raw time variation in remote work across the treated and control groups. As the graph shows, the SDID weights select control units that have a very similar run-up in remote work before the release of ChatGPT, but they differ in their behavior in the post-period that is untargeted by the synthetic control weights. Note that any differences in remote work *levels* between treatment and control groups that can be seen in the graph are eliminated by the firm fixed effects included in the estimation. Panel (B) shows the standard event study plot commonly used in difference-in-differences designs, which illustrates the treatment effect for each period. As the graph shows, there are minimal differences between the treated and control groups in the pre-period, but the estimated gap becomes large and consistently negative starting in the ChatGPT release quarter.

Generative AI effects on the level of job postings. Is the estimated effect on the composition of hiring accompanied by a decline in the *level* of hiring by firms? Columns (2) and (3) of Table IV show the relative decline in total job postings per firm in the treated group after the release of ChatGPT, for both remote and total job postings. The estimates show that overall hiring declines for both remote jobs and for jobs overall, but more than proportionally so for remote jobs. This suggests that generative AI adoption does not merely lead to a reallocation of positions from remote work to non-remote work, but also involves a substitution of the automation tools for human labor overall at the affected firms.

Heterogeneity of effects in tech vs. non-tech firms. We would expect the impact of

generative AI on a firm’s work processes and thus its substitution for other technologies to depend on the nature of the tasks at a firm that are impacted by generative AI—I expand on this idea in the conceptual framework in Section IV. An important distinction highlighted by the sector-level differences in the technology ladder effect shown in Section III.B is that between tech and non-tech firms. Panels B and C of Table IV estimate the effects of GenAI exposure on remote work separately split by whether firms are in the tech sector. The results show that the negative impact on average is driven by a strongly negative effect on remote work within the tech sector (a relative 11 pp decline from a pre-period baseline of a 29.5% remote share), and a small *positive* effect among non-tech firms (a 1.4 pp increase over a 14.9% baseline). The conceptual framework in Section IV, and the additional empirical evidence in Section V suggest that these divergent effects can be explained with the different effects that generative AI has on task composition in tech firms compared to non-tech firms.

Robustness checks. In order to evaluate how robust these SDID results are to changing the specification, I explore the estimates under a number of different variations in Table V. Columns (1) and (2) change the cutoff that defines “high” generative AI exposure to be the 95th or 85th percentile of firms. The results in panel A show that the effect estimates are more negative, the higher (more stringent) the cutoff for high exposure is. The estimates in panel C for non-tech firms show that this is driven by a more stringent cutoff among non-tech firms leading to a less positive estimate—perhaps driven by comparing firms that are more “tech-like” but which are formally outside the tech sector. Panel B shows that coefficient estimates within the tech sector do not vary much across different choices of cutoff. This variation across cutoffs may also arise because exposure is more strongly associated with actual generative AI adoption among highly exposed firms, as shown in Appendix Figure A.5.

Column (3) residualizes the remote share with regard to 2-digit-industry-by-year fixed effects before the SDID estimation, which leads to precise zero effect estimates among non-tech firms and an unchanged estimate among tech firms⁸. I also investigate whether the change in remote work trends for exposed firms around the ChatGPT release may be spurious, by estimating a placebo event study that takes place before the release of the Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser, and Polosukhin (2017) paper that originated the transformer architecture underlying ChatGPT. Here, I define the sample to be the 15 quarters Q4 2013-Q2 2017, and the post-event period as the period Q3 2015-Q2 2017. Column (4) shows precisely estimated null results of this placebo event for the remote work effect of GenAI exposure (measured in 2022). That is, firms matched on remote work pre-

⁸As there are only two 2-digit industries within the broader tech sector (NAICS 51 and 54), we would not expect the estimate to change much within the tech sector.

trends but with different GenAI exposure do not exhibit differential trends after a placebo event the way they do after an event that is related to generative AI technology. This supports the idea that the difference in post-ChatGPT behavior is not due to an unobserved characteristic of firms that correlates with GenAI exposure and might drive differential hiring behavior independent of the ChatGPT release shock.

In summary, the large and negative technology substitution effect in the tech sector is robust to varying the specification, while the small positive effects in the non-tech sector in the baseline specification becomes even smaller and indistinguishable from zero in some specifications. In order to understand *why* remote work and generative AI adoption impact one another in the observed ways and with this pattern of sectoral heterogeneity, I next propose a mechanism that can rationalize these results.

IV. Conceptual framework: organizational technology ladders

Why would having many workers in remote positions *cause* a firm to adopt generative AI at a faster rate? And why would generative AI adoption reduce remote hiring, but only in some firms? In this section, I provide a conceptual framework for how remote work and generative AI adoption can be linked that is based on a firm’s decision to invest in different technologies, and the organizational adaptation to these technologies. I show how one technology might represent a “ladder” to adopting another technology, which introduces path dependence into firm investment choices, and interdependence between the technologies in their impact on firms, workers, and the broader economy.

Model overview. I develop a parsimonious partial equilibrium model of the firm choice of investment in technology and its consequences for organizational structure. The model builds on the task-based approach to modeling automation (Acemoglu and Restrepo, 2022; Thompson and Autor, 2024) but explicitly incorporates firm tech-savviness, investments in new technologies, and variable decision-making sensitivity of tasks. This model is meant to be a simplified illustration of how one can explain the empirical results with regard to technology ladder and technology substitution effects, and is meant to guide the interpretation of the additional evidence on organizational adaptation with regard to skill demand and effect heterogeneity in the following sections. This framework captures all of the mechanisms discussed in Section II.A.

In this setting, I model remote work adoption as a trade-off between greater effective work hours for the worker and less effective support from managers in making effective

decisions. In contrast, Generative AI adoption expands the frontier of tasks that are automated, which may be more or less tilted towards decision-intensive tasks. Sequentially adopted technologies are linked both due to changes in a firm's task composition, and because firms invest in technology skills and information and communication technology (ICT) capital to adopt one technology, which can then be reused for deploying the next one.

Based on the three assumptions that (i) existing ICT capital can be reused for generative AI; (ii) remote work reduces the quality of decision-making; and (ii) generative AI adoption changes the decision-making intensity of human tasks, this framework can explain both the increase in generative AI adoption for firms that go remote, and why it leads, in some sectors, to a reduction in remote hiring for firms that adopt generative AI. One of the key mechanisms is that the change in the decision-making intensity of human tasks changes the productivity of remote work. Importantly, the qualitative predictions of this framework suggest empirical tests to validate the proposed organizational adaptation mechanism.

A. Occupational output

For simplicity, assume that each firm consists only of one production occupation and one production worker, with potentially some non-producing “managers”—although the results would be the same if a firm has multiple workers in the same occupation. A worker at firm f in occupation j , which, in this simplified framework also indexes the type of firm that employs the worker, produces output by executing the non-automated subset of the tasks $x \in [0, j]$ that are sorted by automatability. Output y_{fj} is given by the following expression:

$$\ln y_{fj} = \underbrace{k_{fj}}_{\text{Automated production}} + \underbrace{\int_{k_{fj}}^j \ln\left(\frac{1}{j - k_{fj}}\right) dx}_{\text{Non-decision tasks}} + \underbrace{\int_{k_{fj}}^j \delta_f(x) Q_f dx}_{\text{Decision-making}} + \underbrace{\int_{k_{fj}}^j r_j \times \mathbb{I}[\text{Remote}_f] dx}_{\text{Remote time savings}} \quad (3)$$

Here, the first term is the automated product $k_{fj} < j$, which represents how much of the task spectrum $[0, j]$ is automated (or does not require labor) at firm f . In the second term, the integral $\int_{k_j}^j$ can be seen as a stylized representation of the “remaining tasks” from k_j to j that must be performed by labor of firm f .

The third term can be thought of as either a productivity “boost” from good decision-making or as the output from a separate planning task that accompanies each part of the worker's job, and which varies in importance across occupations. Its output depends on two factors: The term $\delta_f(x)$ captures the importance of decision-making for tasks of different complexity that the worker does. As the tasks are sorted by automatability, the slope of $\delta_f(x)$ determines whether the tasks that are automated first are relatively low or high in

decision-making intensity. Consider, for example, the stylized illustration in panel (a) of Figure 4: it shows automation for tech firms as having relatively low “judgment skew”: decision-making intensity is lower for tasks that are more likely to be automated. This means that the remaining human tasks become more decision-intensive with automation. In contrast, the graph shows non-tech firms as having a flat distribution of decision-making intensity with regard to automatibility, indicating that automation has no judgment skew. While this is only a stylized example, it anticipates the across-sector effect patterns that I find in the mechanism-related estimations below. For tractability, assume

$$\delta(x) = e^{\lambda_s x},$$

where sectors s with a higher λ_s have a lower judgment skew of automation, as decision-making intensity increases with distance from the automation frontier.

The other factor driving the decision-making output is Q_f , which is the worker’s decision-making quality. Decision quality is defined as

$$Q_f = \ln(\rho_f^{\mathbb{1}[\text{Remote}_f]} M_f),$$

where M_f is the decision support from the firm (e.g. from better central management or technology infrastructure), and $\rho_f < 1$ represents a barrier to communication and coordination that reduces the effectiveness of this decision support. Here, I explicitly make this barrier a function of remote work, but it could also be used to model the coordination effects of a firm’s technology choices more generally, e.g. when thinking about offshoring or outsourcing. Moreover, ρ_f can vary across firms, as some firms having better leadership or communication skills that can bridge the barrier and therefore have a higher ρ_f (a smaller discount factor on decision support when remote).

The last term in equation 3 represents the productivity benefit of working remotely, which may differ by occupation. This allows for some occupations where working from home is easily accomplished and productive, while in other occupations it may either be physically infeasible (e.g. for an emergency room nurse) or undesirable for other reasons. It is a function of r_j , the remote suitability of the occupation, and an indicator $\mathbb{1}[\text{Remote}_f]$ for whether the firm is remote.

B. Technology adoption

The firm decides in each period t whether to adopt a new technology τ , if one is available. Technologies are assumed to arrive at random and new technologies are not anticipated by

the firm. The firm’s technology status—the last technology it adopted—when making its period t adoption decision is τ_f .

Technology implementation costs. Each new technology τ that the firm adopts—where, for concreteness, I will focus on $\tau \in \{\text{Remote}, \text{GenAI}\}$ —requires updating the firm’s level of information technology to $I(\tau)$, which has cost

$$c_f(\tau) = \frac{1}{T_f} (I(\tau) - I(\tau_f))^2,$$

where T_f is a measure of the firm’s innovation orientation or tech-savviness, which reduces its investment costs. That is, new investment is costlier if the firm did not adopt previous technologies, such that there is a bigger technology gap $I(\tau) - I(\tau_f)$ between the firm’s level of technology infrastructure and the level required for the new technology. I assume that later technologies always require a higher level of information technology, and that remote work is the last technology wave that precedes generative AI.

Optimal technology adoption. Each firm adopts a new technology if the benefits of doing so outweigh the benefits *or* if the firm is forced to do so by policy, for example, having to adopt remote work as a result of pandemic lockdowns. Generally, the firm adopts new technologies if the increase in output is greater than the cost of adopting the technology, i.e.

$$\ln y_{fj}(\tau) - \ln y_{fj}(\tau_f) > c_f^\tau.$$

Note that a firm of a given type j will be more likely to adopt a given technology if it is more tech-savvy (higher T_f), or if it recently invested in other advanced technology, i.e. when the technology gap $I(\tau) - I(\tau_f)$ is small, i.e. if reusability of ICT is high.

Different technologies will change different aspects of the production technology, potentially affecting the automation of tasks, the benefits from remote work, the cost of communicating with managers, etc. I will focus on a particular sequence of technology waves that reflects the large changes experienced during the 2020-2024 period: First, a change in the attractiveness of remote work technology as a result of the Covid pandemic. Second, a change in the potential for automation of particular tasks as a result of the proliferation of generative AI based tools.

C. Technology Wave #1: Remote work

While there were certainly firms that allowed for remote work before the Covid pandemic in 2020, the sudden need to continue work when forced to work-from-home during lockdowns led to both a large increase in the adoption of remote work in “teleworkable” oc-

cupations, as well as large investments, and rapid improvements, in the technology tools enabling work-from-home, e.g. virtual meeting and collaboration software.

Here, remote work is assumed to provide an additional productivity boost r_j to all the tasks done in an occupation, as shown in equation 3. This is motivated by empirical observations that remote workers save time on grooming and commuting, and have more flexibility in their time use over the day, and that some of these time savings are used to work more (Barrero, Bloom, and Davis, 2020; Pablonia and Vernon, 2022). This productivity boost can vary across occupations, such that there might be “teleworkable” jobs with a high r_j and other jobs where r_j is small or even negative (Dingel and Neiman, 2020). At the same time, remote work reduces the quality of decision-making support from management through triggering the communication barrier $\rho_f^{\mathbb{1}[\text{Remote}]} < 1$. This assumption is empirically supported by the findings in Choudhury, Espinosa, Makridis, Schirmann, et al. (2025) that workers coordinate less with managers when they are not co-located.

Based on the model above, a firm will adopt remote work (indexed by R) rather than stay non-remote (NR) if the overall output increases as a result. The trade-off is that remote work decreases overall decision quality, which represents a cost to be weighed against any potential productivity benefits.⁹

Empirical predictions. This model predicts particular patterns of remote work adoption among firms:

- P1. More teleworkable occupations and firms (higher remote work benefit r_j) are more likely to adopt remote work.
- P2. More tech-savvy or innovative firms (higher T_f) are more likely to adopt remote work for a given occupation.
- P3. Firms with a higher level of existing technology investments (a smaller technology gap $I(\tau) - I(\tau_f)$) are more likely to adopt remote work
- P4. Higher adoption of remote work leads to an increase in demand for managerial skills and technology skills in the firm’s workers.

Some of these predictions are already supported by existing findings in the remote work literature, e.g. the teleworkability measure by Dingel and Neiman (2020) has been shown to predict at least part of the variation in actual remote work adoption (Hansen, Lambert, Bloom, Davis, Sadun, and Taska, 2023), which confirms P1. In the empirical sections of this paper, I provide novel evidence for predictions P2, P3, and P4.¹⁰

⁹See Appendix Section D.A for details.

¹⁰Specifically, see Section C.A for evidence on P2 and P3, and Section V.A for evidence on P4.

D. Technology Wave #2: Generative AI and the technology ladder effect

I model the effect of generative AI tool availability as pushing the automation boundary k_j upward. That is, if tasks are sorted by j , then the frontier of automatable tasks expands to

$$k_{fj}^G = k_{fj} + \underbrace{\text{GenAIPotential}_j \times \mathbb{1}[\text{GenAIAdoption}_f]}_{\Delta k_{fj} \text{ with Gen. AI}},$$

where G indexes values conditional on the adoption of generative AI. Here, $\text{GenAIPotential}_j \geq 0$ measures how amenable occupation j is to the new technology, i.e. what share of its tasks are exposed to generative AI automation. Whether this exposure leads to automation at the firm depends on $\text{GenAIAdoption}_f \in \{0, 1\}$ which indicates whether a firm invests in generative AI adoption. If a firm invests in the necessary information technology capital $I(\tau = G)$, it moves more tasks into the automatable set. Hence, the fraction of tasks requiring actual worker effort shrinks from $[k_{fj}, j]$ to $[k_{fj}^G, j]$. Note that when k_j increases, more tasks become automated, so fewer tasks remain for labor—but if $\delta'_f(x) > 0$ (i.e. $\lambda_f > 0$) the tasks that remain benefit more from higher-level decision-making. See Figure 4 for an illustration: the diagonally hatched area represents the tasks that are automated as the automation frontier moves from k to k^G , and for tech firms, which have an upward-sloping decision-intensity curve in this stylized example, this means that the tasks that remain for humans post-automation are more decision-intensive.

The firm adopts generative AI if $\ln y_{fj}^G - \ln y_{fj}^{\text{NG}} > c_f^G$. Note that the cost c_f^G is lower for firms with high existing technology capabilities, so this directly implies another prediction which I can confirm in the data:

P5. Firms with higher levels of technology skills are more likely to adopt generative AI.

Remote work effect on generative AI adoption. How does the adoption of remote work affect the decision of whether to adopt generative AI technology in this model? Remote work changes worker productivity, which will determine the productivity loss or gain when generative AI is used to automate a remote human worker compared to a non-remote worker. At the same time, the cost of investing in generative AI will be lower if the firm previously adopted remote work, as firms that adopt remote work have a higher level of information technology capital than those which do not. Intuitively, the trade-off is that the work quality of remote work is lower, so replacing it with generative AI is more profitable, and remote firms also already have some of the necessary infrastructure in place—but this has to be weighed against remote workers' longer effective work hours.¹¹

¹¹See details in Appendix Section D.B.

If we make the assumption that the work-from-home productivity boost for the automatable tasks is not too large relative to the technology investment cost savings from generative AI adoption, this model can rationalize a **technology ladder effect**:

P6. *Firms that adopt remote work have a higher incentive to also adopt generative AI if the reusability of ICT capital is high.*

Note that it is an empirical question whether the underlying assumption is realistic. Showing that this effect exists is one of the key contributions in the empirical section of this paper. There may be heterogeneity in this effect across firms: some organizations are better able to manage remote work and keep the decision quality loss from going remote small, which then reduces the relative benefits of generative AI adoption between remote and non-remote firms. Conversely, some firms have a lower remote effectiveness ρ_f . Assuming that firms that public issue “return-to-office” mandates are thereby revealing that they perceive the productivity of remote work to be low, the model predicts the following heterogeneity for the technology ladder effect:

P7. *The effect of remote work on generative AI adoption is stronger for firms with return-to-office mandates.*

Generative AI effects on remote work. Conversely, does generative AI adoption impact a firm’s incentive to continue using remote workers in this model? Intuitively, the average decision-making sensitivity of tasks likely increases with automation, as lower-complexity tasks are more likely to be automated. As decision-making quality is worse in remote work, the overall benefit from remote work therefore declines with greater automation. However, there may be substantial heterogeneity in this effect across different sectors. This means that the model predicts a **technology substitution effect**:

P8. *Higher adoption of generative AI leads to a greater decline (or smaller increase) in the use of remote work technology for firms if automation has a low judgment skew, i.e. $\lambda_s > 0$.*¹²

For example, if $\lambda_{\text{Tech}} > \lambda_{\text{Non-tech}}$, i.e. generative AI is more likely to automate relatively low decision-making tasks in the tech sector, we would expect the effect of generative AI exposure on remote work to be more negative in the tech sector. The size of this effect of generative AI adoption on remote work also depends on the decision quality loss due to

¹²Note that I am assuming that for the purpose of the decision of whether to continue letting workers work remotely, the technical infrastructure is not relevant, as there is no way to “recoup” the original outlay for information technology capital to enable work-from-home when later eliminating remote work.

remote work adoption. This means that the same factors that reduce the impact of remote work on decision quality will also reduce the technology substitution effect. Assuming that automation on average has a low judgment skew ($\lambda_s > 0$), consistent with the empirical results that I find, it should vary with firm characteristics as follows:

P9. *Adoption of generative AI leads to a smaller decline in the use of remote work for firms with greater communication, managerial or leadership skills.*

In the next section, I will further explore the empirical evidence to see whether it aligns with the predictions from this conceptual framework that go beyond the technology ladder effect and technology substitution effect that are supported by the results in Sections III.B and III.D.

V. Mechanism: organizational adaptation

The empirical results in in Sections III.B and III.D showed that greater remote work adoption leads companies to embrace generative AI, while the adoption of generative AI is followed by a decline in remote hiring in the tech sector. The conceptual framework proposed in Section IV suggests that these patterns can be explained with a model where workers' quality of decision-making declines when they work remotely, and the use of generative AI tools tends to increase the importance of strategic decision-making on the part of the human worker. In this section, I provide evidence on how empirical adoption patterns in different organizations and changes in hiring for particular skills align with the proposed mechanism.

A. *Building the organizational technology ladder: remote work and hiring*

This section tests how the adoption of remote work changed firm-level hiring patterns with regard to skills. These results help us understand through which mechanism the technology ladder effects on subsequent generative AI adoption might be operating. I estimate specifications of the form

$$\text{Skill}(2022)_i = \alpha + \beta \text{RemoteWorkShare}('21-'22)_i + \gamma \text{Skill}(2019)_i + \text{Controls}_i + \varepsilon_i, \quad (4)$$

where the dependent variable captures different measures of the composition of the firm's hiring with regard to indicators of skill in its job postings, or the average characteristics (derived from O*Net scores) of the occupations that the firm is hiring for. These regressions also control for the past level of the dependent variable pre-pandemic, so the estimated

coefficients capture the effect on changes in the skill composition of job postings between the period before and after the lockdown period of the pandemic.

Figure 5 shows the results of the IV estimation, which additionally controls for a rich set of pre-pandemic hiring characteristics to capture potential confounders in hiring skill trends.¹³

First, I explore whether firms change the complexity of the positions they hire for, by looking at changes in the formal requirements for workers’ experience or education, and in the repetitiveness of tasks, when they are increasing the remote share of their workers. Section A of Figure 5 shows that firms that adopt remote work at a higher rate increase their hiring of workers with at least 5 years of experience, and also hire for less repetitive roles, but do not hire significantly more for advanced degree holders.

Second, I explore whether firms change their hiring composition in ways that might affect the nature of work processes within the firm: Section B of Figure 5 shows that firms hiring remotely do not significantly shift their overall hiring towards roles that tend to involve more interactiveness or “inflexible” client responsibilities as defined by (Goldin, 2014). However, I do find evidence of more hiring for roles requiring greater social skills as defined by Deming (2017) when remote work is adopted. Section C of Figure 5 shows a significant shift in firm hiring towards roles having greater Deming (2021) decision-making intensity, i.e. *requiring* greater individual decision-making, as a result of remote hiring.

Third, I estimate the effect of remote hiring on firm demand for particular skills that might serve to adapt the organization to better enable remote work. One of the largest shifts in the composition of hiring among the categories considered is towards roles with technical skills: the share of computer occupations and of job postings mentioning data management, data science, machine learning, or deep learning skills increases significantly in firms where remote work increases, as shown in section D of Figure 5. This suggests that firms are investing in their digital infrastructure and related skills when they have more remote workers, in line with the proposed technology ladder channel operating through greater IT capital. Section E of Figure 5 shows that more remote firms also hire more for roles that involve greater leadership skills, and for more managerial roles. This suggests that firms may increase their demand for centralized managerial skills to deal with the increased difficulty of supervising remote workflows.

¹³The firm-level regressions include the following control variables: NAICS 2-digit fixed effects; company’s remote work share in 2019, the company’s uninteracted exposure to MSA teleworkability in 2019, uninteracted firm-level teleworkability in 2019, the company’s share of jobs requiring a college education and the share requiring an advanced degree in 2019, the share of the company’s job postings in 2019 that was for computer occupations or manager positions; the log of total job postings in 2019 and in 2022; the company’s labor market exposure to MSA remote shares in 2019.

Firm characteristics and generative AI adoption. The proposed technology ladder mechanism also requires that these changes in the composition of hiring and skills make it easier for the firm to adopt generative AI technology. In Appendix C, I provide evidence that firm-level generative AI exposure is indeed more likely to result in actual generative AI adoption in firms with the characteristics that the adaptation to remote work reinforces. Appendix Figures A.9 and A.10 show that firms hiring for above-median levels of tech skills, social skills, lack of repetitiveness, education, experience, managerial roles, and decision-making intensive jobs are indeed more likely to translate a given level of exposure to predicted benefits from the technology into actual adoption. The largest differential effects are for firms that have high levels of investment in technology skills, in line with the idea in the conceptual framework that higher technology skills should make generative AI adoption more likely. Together with the evidence on the changes in skill composition in response to remote work adoption, this provides the second part of the technology ladder mechanism: changes in firms’ hiring composition in response to remote work skew hiring towards exactly those skills that make it more likely that a firm adopts generative AI when there are potential benefits from the technology.

B. Heterogeneity of remote work effects on generative AI adoption

In this section, I estimate whether particular firm characteristics are associated with higher or lower impacts of remote work adoption on generative AI adoption (in line with predictions P7 and P8 in Section IV). I estimate interacted regression models of the form

$$100 \times \text{GenAIJobShare}('23-'34)_i = \beta \text{RemoteWorkShare}('21-'22)_i + \alpha_{ind} + \text{Controls}_i \\ + \gamma \mathbb{1}[\text{High}_f] \times \text{RemoteWorkShare}('21-'22)_i + \varepsilon_i$$

where the “High” indicator is computed for different firm characteristics and indicates that the firm is above the median across firms for the measure as of 2022. The IV estimations use the same instruments as in the baseline estimates in Table II, as well as an interaction between the instrument and the group indicator.

The results of estimating these interacted models are shown in Figure 6, where I show only the coefficients on the interaction term that captures the difference in remote work effects between the ‘high’ and the ‘low’ group for each characteristic. Panel A shows the estimates for the across-firm estimation and panel B for the within-firm estimation (corresponding to heterogeneity in the baseline estimates in columns 2 and 4 of Table II). Importantly, Appendix Figure A.11 shows the corresponding ‘low’-group effects, which are consistently positive, showing that the baseline technology ladder effect is not driven by a particular

subset of the data based on the characteristics considered.

I find patterns that align with the proposed conceptual framework of technology adoption: there is systematic variation in the size of the effect with particular firm characteristics: (A) firms with greater advanced degree hiring and less repetitive jobs are more likely to adopt generative AI when workers are remote, and also show greater responsiveness of adoption to remote shares within the firm; (B) firms that hire for occupations that require more social skills, are interaction-intensive or inflexible are not overall more responsive in their generative AI adoption, but are more likely to adopt generative AI in more remote roles within the firm; (C) Firms that are more decision-intensive are more likely to deploy generative AI into roles with greater remote work shares within the firm, but are not overall more responsive; (D) firms that invest in specialized tech skills—in particular, data science, machine learning, and deep learning, are significantly more responsive to remote work in adopting generative AI overall, but do not have differential responsiveness in targeting generative AI adoption to more remote roles within the firm; and (E) firms hiring for more manager roles are more likely to adopt generative AI after going remote, and also show greater within-firm elasticity of generative AI adoption to remote work.

These findings directly support the notion that organizational adaptation matters for the ability to deploy technologies: Across firms, those that have more complex positions (more advanced degrees and repetitiveness), more managerial capacity, and invested in greater tech skills are more likely to invest in generative AI in response to deploying remote work during the pandemic. Within firms, generative AI is more likely to be deployed specifically in high remote work roles when the firm’s job composition is more complex, if the firm has high communication intensity or decision intensity, or greater managerial skills.

C. Return-to-office policies and generative AI adoption

The previous results suggest that greater generative AI adoption in response to remote work may be associated with particular organizational characteristics, such as complex tasks, that may make remote work harder to sustain, or with characteristics, such as tech skills, that make it easier to implement the new technology.

In this section, I further explore the idea that not just the technological capability to adopt generative AI matters, but also the adaptation to the previous remote work wave—and the firm’s productivity while working remotely. To identify this variation in adaptation, I consider firms that have publicly shown that remote work is not working out for them by issuing “return-to-office” mandates. I use data on firms that have stated RTO policies, collected from the crowd-sourced Flex Index website. I create an indicator for whether a

firm appears in the Flex Index data as requiring a minimum presence in the office. All other firms (whether they appear in Flex Index or not) are labeled as not having an RTO mandate.¹⁴

Then, I estimate IV regression specifications of the form

$$100 \times \text{GenAIJobShare}('23-'34)_i = \beta \text{RemoteWorkShare}('21-'22)_i + FE_s + \text{Controls}_i \\ + \gamma \mathbb{1}[\text{RTOMandate}_i] \times \text{RemoteWorkShare}('21-'22)_i + \varepsilon_i$$

both at the firm and at the occupation-by-firm level. For the purposes of the estimation, I treat RTO mandate status as exogenously assigned, conditional on the control variables, which are the same as in the baseline effect estimations in Table II. The results are shown in Table VII: I find that firms that have an RTO mandate are more than twice as likely to adopt generative AI for each percentage point of remote hiring during the pandemic (column (1)). Moreover, in line with the idea that this remote work aversion is based on a firm-level lack of adaptation to remote work leading to a firm-wide investment in generative AI, rather than a targeted adoption only in remote jobs, column (2) shows that there is no evidence that the within-firm effect of remote work on generative AI is different in RTO firms.

D. Technology substitution effect heterogeneity by firm characteristics.

Which firm characteristics make it more likely that a firm responds to generative AI adoption with a reduction in remote hiring? I split the sample based on the characteristics of a firm’s hiring in 2022 and repeat the synthetic diff-in-diff estimation for each subgroup. The results are shown in Figure 7: Comparing the above-median ‘high’ group estimates to the ‘low’ group in each category, some clear patterns emerge: (A) Among firms with more experienced workers and less repetitive tasks, generative AI exposed firms are more likely to reduce remote hiring after the ChatGPT release; (B & C) This negative effect only exists among low communication intensity and high decision intensity firms and is reversed (i.e. remote work increases) among high communication intensity and low communication intensity firms; (D) Among firms with greater technology skills the technology substitution effect is stronger (more negative); and (E) among firms with high managerial capacity, generative AI adoption is associated with more remote hiring, while remote hiring declines with generative AI among low managerial skill firms.

This heterogeneity with regard to decision-making skills can be interpreted in the context of the mechanism captured in the model in Section IV: firms that have more jobs with

¹⁴See Appendix B for details on the Flex Index data.

high decision intensity, few repetitive tasks, and which hire advanced degree workers are likely to have a low judgment skew in automation, such that generative AI automation increases the average decision intensity of human tasks, and remote work becomes less productive. The evidence also suggests that generative AI adoption may actually reduce the remote work penalty for communication-intensive firms. This may be due to generative AI automation having high judgment skew for communication-intensive jobs: for example, generative AI may take over planning and customization tasks in communication and teaching jobs, but leaves the execution tasks—such as presentations, calls, and emails—to human workers. The results related to tech skills and management skills show the role of organizational capabilities: among firms with greater ability to implement generative AI (high tech skills), generative AI adoption may be more effective and therefore replaces remote work to a larger degree. However, if firms have high managerial skills, remote work penalties may be smaller, reducing the pressure to reduce remote hiring when generative AI is adopted.

How do the technology substitution effects vary across sectors? Table IV already showed that the effect is large and negative in the tech sector and small and positive in non-tech sectors. Appendix Figure A.12 breaks the effect down further, estimating the SDID effects within broad sectors of the economy. The results show that effects are particularly positive in health care, social assistance, education, and other service sectors, and more negative in tech, but also in business support and financial services.

The next section provides additional evidence of changes in job characteristics as a result of generative AI that support the notion of a low judgment skew of generative AI automation particularly in the tech sector.

E. Hiring composition and the technology substitution mechanism

What happens to the skill composition of hiring at firms with high exposure to generative AI after ChatGPT is released? I repeat the same estimation as in equation 2, but with the share of remote jobs or non-remote jobs that have particular characteristics as the dependent variable. This analysis captures whether the exposure to generative AI further changes the skill mix of a firm’s overall hiring after the ChatGPT release—and if these effects differ between tech and non-tech firms.

The effects of generative AI exposure on skill composition are shown in Table VI: In panel C, columns (1) and (2) show that exposed non-tech firms hire more for high experience workers, but less for advanced degrees after the ChatGPT release, in their non-remote jobs, with no significant change among remote jobs in columns (6) and (7). However, tech firms in panel B show no evidence of changes in demand for formal education or experience.

Columns (3)-(5) and (8)-(10) then focus on specific skills being mentioned in job postings: Columns (3) and (8) shows that there is some evidence that non-remote non-tech jobs may be lowering their demand for data management skills, but data management skill demand does not shift substantially in other groups. Columns (5) and (10) show that demand for communication skills increases at exposed firms in remote jobs in the tech sector, but not in non-remote jobs or outside the tech sector.

Importantly, this analysis allows me to estimate the empirical equivalent of the notion of “judgment skew” of generative AI automation: Column (4) shows a significant effect on demand for decision-making skills in the tech sector among generative AI-exposed firms—and this effect is large and positive both for non-remote and remote jobs. In the non-tech sector, this effect is negative but not statistically significant. This suggests that jobs in the tech sector become more decision-intensive in exposed firms after ChatGPT is released—consistent with the stylized “low judgment skew” automation example shown for the tech sector in Figure 4.

VI. Discussion

The results above tell the following story of how the technology ladder between remote work adoption and generative AI adoption operates: First, many firms adopted remote work in response to greater availability of related tools and a physical need to adopt them during the Covid pandemic, which then changed optimal work processes and firms’ skill mix.

Adaptation to remote work. I find that the change in work processes that resulted from this first technology shock was associated with hiring more experienced workers, consistent with the finding by Emanuel and Harrington (2024) that the productivity effects of remote work are more negative for less experienced workers. I also find evidence that remote work caused firms to hire for skills that would allow them to better adapt to the new work style by hiring additional managers. This is consistent with the increased demands on managers as a result of a reduced ability to effectively train and supervise remote workers.

Demand for technical skills related to information processing, such as machine learning, also increases substantially in more remote firms. This is in line with other studies that have found that firms increased the share of investments going to IT equipment and computers during the pandemic (Barrero, Bloom, and Davis, 2021), and suggests that they are investing in broader digital capabilities and technical infrastructure to support remote workers. Additionally, I find some evidence that firms increased their hiring for social skill-intensive roles. Other research has found that remote work increases the time and effort

devoted to communication (Gibbs et al., 2023) and leads to a decrease in the amount of information sharing across collaboration networks (Yang, Holtz, Jaffe, Suri, Sinha, Weston, Joyce, Shah, Sherman, Hecht, et al., 2022), which firms might seek to counter-act by investing in hiring for management and team coordination roles that can overcome these issues.

Technology ladder effects. After ChatGPT was released, remote work then facilitated the adoption of generative AI. While the causal effect on generative AI adoption is robust across different sub-samples, the heterogeneity tests reveal that technology skills in particular differentiate which firms implemented generative AI at a faster pace in response to initially going remote. Together with the evidence on changes in hiring as a result of remote work, this suggests a direct mechanism through which organizational adaptation increases the benefits from the subsequent generative AI technology wave: firms invest in technology skills to accommodate remote work; generative AI adoption then builds on this technical infrastructure and the skill investments made for remote work.

Technology substitution. While remote work adoption enables the use of generative AI tools, the use of the latter also displaces hiring for the former. That is, the event study design around the release of ChatGPT shows that firms that are more exposed to generative AI reduce their hiring for remote workers. But the size of this effect depends on an organization’s ability to accommodate remote work: the same non-technology characteristics—such as hiring more for social skill-intense roles or for managerial skills—that enable the adoption of generative AI in response to remote shares also make it less likely that generative AI exposure leads to a reduction in remote hiring. Moreover, the negative effects on remote work are concentrated in the tech sector, while remote work does not decline substantially more in generative AI exposed firms in the non-tech sector.

More generally, this suggests that firms that incur a smaller quality penalty from going remote are less likely to demand generative AI automation, in line with the latter representing a substitute for “bad” remote work. In contrast, firms which more actively mitigate the management and coordination issues arising from remote work are less eager to adopt generative AI. A salient example of this dynamic are firms which have issued return-to-office mandates to overcome the perceived disadvantages of remote work for their organizations: these firms see a much larger adoption of generative AI tools relative to their remote hiring than other firms.

Judgment skew of GenAI automation. Finally, organizations also change as a result of generative AI exposure: notably, exposed firms hire more for decision-making skills in

the tech sector, suggesting that generative AI increases the decision intensity of the tasks done by humans. The conceptual framework illustrates how this “low judgment skew” of automation in the tech sector can rationalize the larger technology substitution effect as a result of generative AI exposure among tech firms: if remote work makes it harder to productively work in decision-intensive roles, this increase in decision intensity of human work with generative AI would increase the productivity penalty for remote workers. In contrast, in other sectors where generative AI shifts human labor towards less decision-intensive tasks, the productivity of remote work does not change.

Implications. The dynamic of this “organizational technology ladder” raises important issues for policymakers and businesses responding to technological changes. It suggests that differences in the ability to innovate and to transform an organization in response to an initial technology shock can compound into broader competitive advantages over time if the ability to benefit from later technological waves depends on how eagerly an organization embraced the former. This path dependency means that an initial heterogeneity in technology exposure can cascade into some firms and worker groups being systematically affected by later changes in a way that would be difficult to anticipate based on studying these technology shocks in isolation. As I argue in this study, remote work and generative AI technologies are intimately connected in their effect on firms and local labor markets, both by coinciding in the same jobs and by mutually affecting firms’ ability and willingness to invest in the respective other technology.

As both technologies continue to evolve and firms continue to adapt, understanding these technological complementarities is crucial for predicting how labor markets and firm productivity will evolve, and for policymakers and researchers to be able to design policies that harness these changes for the benefit of society. In addition, the effects on skill demand and heterogeneity of effects for different roles highlight that particular worker groups—particularly those in interactive or technical roles, or in organizations with better communication and managerial capabilities, or who have good decision-making skill—might be less negatively impacted by generative AI, even if their position is remote. My results suggest that transforming firms and educational pathways to prepare workers and employers to take advantage of the current technological transformation may even prepare them better for the next wave of technology to come along—whatever that may be.

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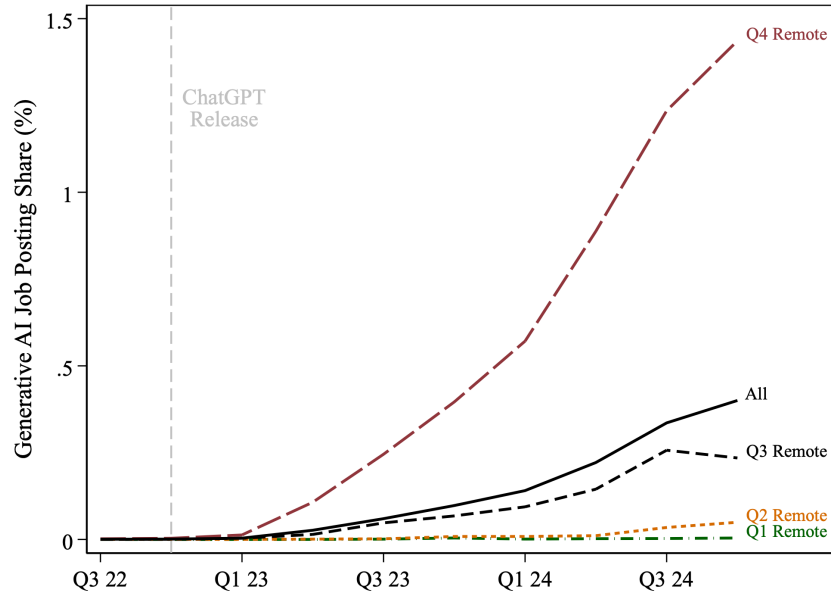
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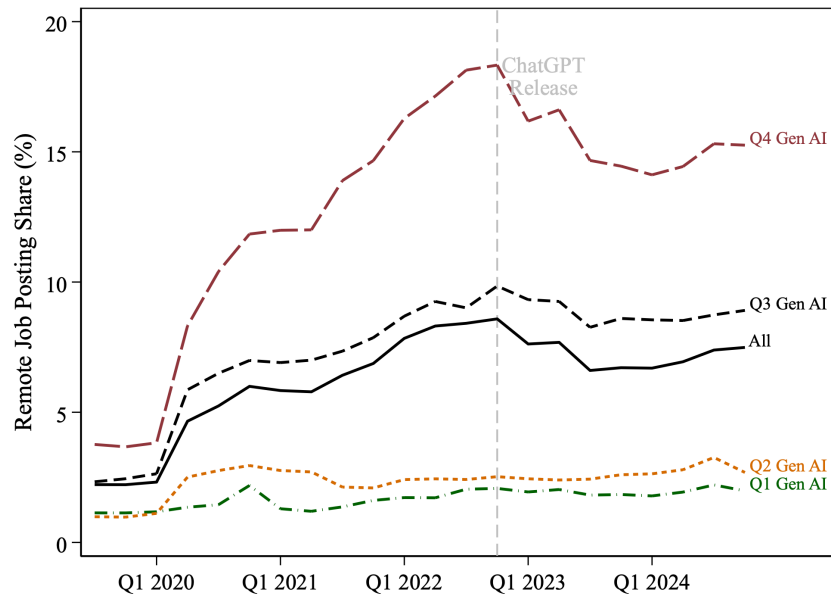
Figures and Tables

Figure 1:
Remote work and occupational generative AI adoption trends.

This figure shows the share of job postings in each quarter that are for jobs that mention generative AI (panel A) or for remote jobs (panel B). In each panel, the occupations are aggregated into job posting-weighted quartiles of adoption of the other technology: panel A shows generative AI adoption by the occupation's quartile of national remote work adoption in 2021-2022 (excl. Q4 2022), and panel B shows remote work share by generative AI demand as of YTD Sep. 2024. The grey drop line indicates the period (Q4 2022) when ChatGPT was released.



(A) Gen. AI share by remote work quartile



(B) Remote work share by Gen. AI adoption quartile

Figure 2:
Firm-level and occupation-by-firm level IV first stage

This figure shows binscatters of the first-stage relationships underlying the IV results in Table II. The first panel shows the effect of the interaction between a firm's exposure to MSA level teleworkability and firm-level teleworkability (both measured in 2019) on the horizontal axis, and the firm's job posting remote work prevalence in the YTD as of Sep. 2024 on the vertical axis. The second panel shows the effect of the interaction between a firm's exposure to MSA level teleworkability, firm-level teleworkability, and the occupation-level national hybrid work share in 2021-2022 on the horizontal axis, and the occupation-by-firm's job posting remote work prevalence in the YTD as of Sep. 2024 on the vertical axis. The values on both axes in both graphs are residualized with regard to the same control variables as the IV results in the tables.

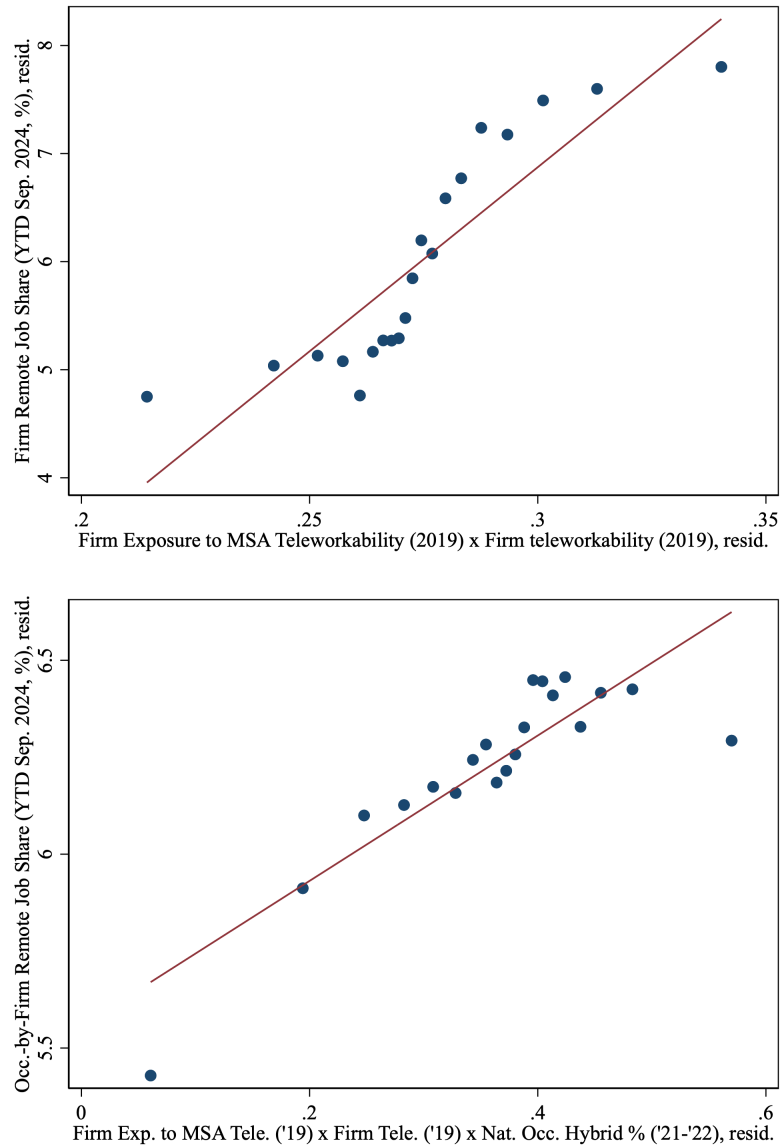
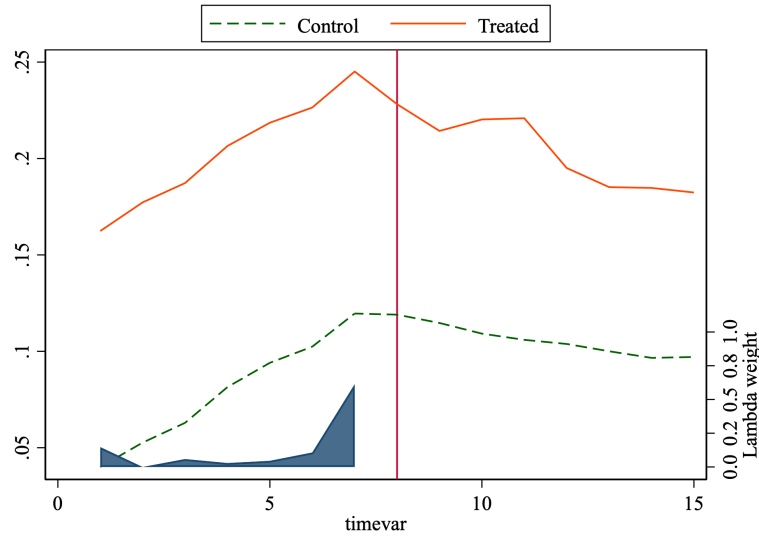
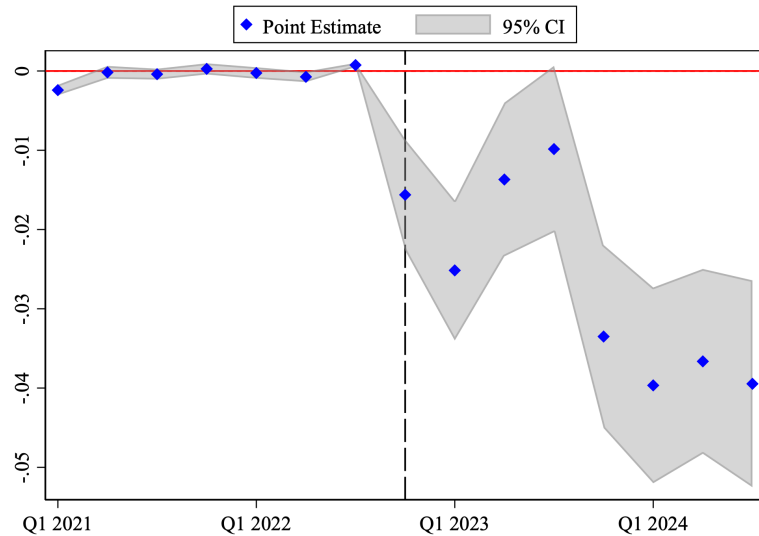


Figure 3:
Synthetic diff-in-diff event study results

These graphs show the differences in remote work share trends between the treated and control groups that underlie the synthetic diff-in-diff effect estimates reported in Table IV. Panel A shows the remote work trends over time in the synthetic control group and the treatment group. The treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). The right axis and the solid area in the graph shows the optimal time weights applied to different pre-periods in the estimation. Panel B shows the estimated difference in remote work shares between the control and treatment group in each quarter (blue dots), as well as the 95% confidence interval based on block bootstrap standard errors computed from 100 resamples, using the procedure detailed in Clarke et al. (2023).



(A) Remote Work Trends: Synthetic Control vs. Treatment Group



(B) Dynamic Treatment Effects of Top Decile Gen. AI Exposure

Figure 4:
Illustration of organizational adaptation dynamics

Panel (a) shows tech vs non-tech decision-making intensity $\delta(x)$ plotted over tasks sorted by generative AI automatability (from high on the left to low on the right), and illustrates the automation frontier shifting from $\mathbf{k}$ to $\mathbf{k}^G$. The hatched area indicates tasks newly automated (between k and k^G). Light shaded area indicates tasks remaining after generative AI automation.

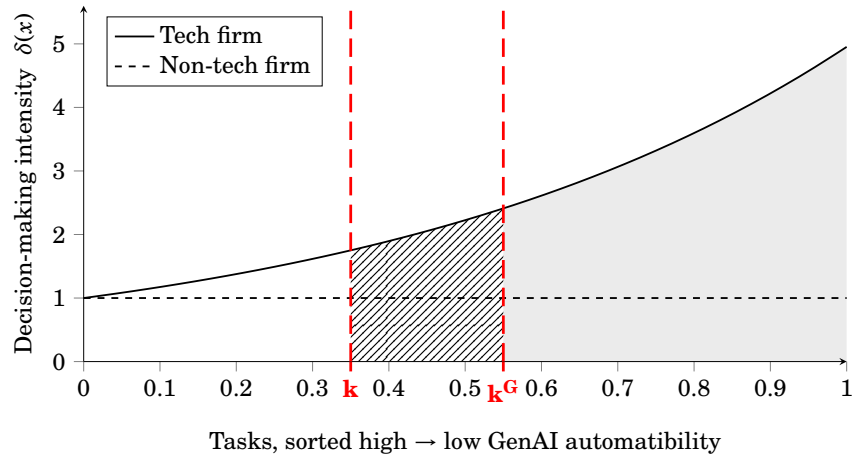


Figure 5:
Effects of remote work adoption on firm hiring characteristics

This figure shows coefficients estimated using IV for the effect of remote work prevalence in a firm's 2021/2022 (excl. Q4 2022) job postings on the standardized characteristics of the firm's job postings in 2022 in a regression of the form

$$\text{Skill}(2022)_i = \alpha + \beta \text{RemoteWorkShare}('21-'22)_i + \gamma \text{Skill}(2019)_i + \text{Controls}_i + \varepsilon_i,$$

where the controls in all regressions include the standardized 2018-2020 average value of the dependent variable as a control variable, so the coefficients can be interpreted as the effect of changes in remote work shares on changes in the composition. The instrument consists of the interaction between firm-level exposure to MSA teleworkability through its hiring labor markets and firm-level teleworkability (all measured in 2019). All regressions also include the following control variables: NAICS 2-digit fixed effects; company's remote work share in 2019, the company's uninteracted exposure to MSA teleworkability in 2019, uninteracted firm-level teleworkability in 2019, the company's share of jobs requiring a college education and the share requiring an advanced degree in 2019, the share of the company's job postings in 2019 that was for computer occupations or manager positions; the log of total job postings in 2019 and in 2022; the company's labor market exposure to MSA remote shares in 2019. The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the NAICS 2-digit sector level.

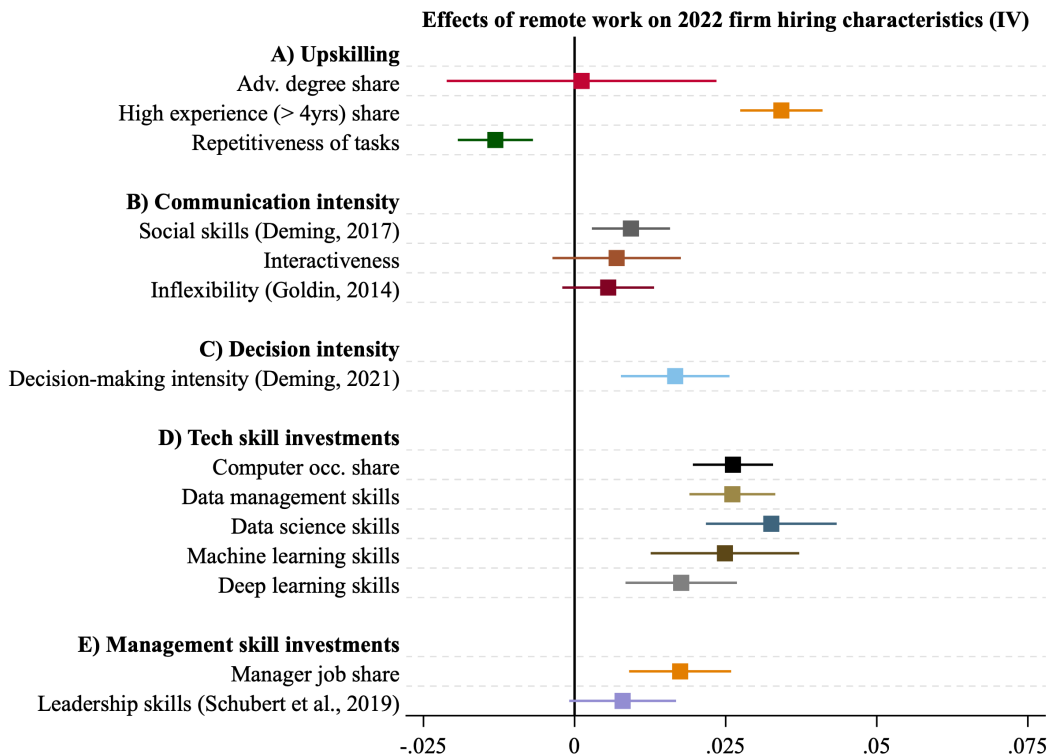
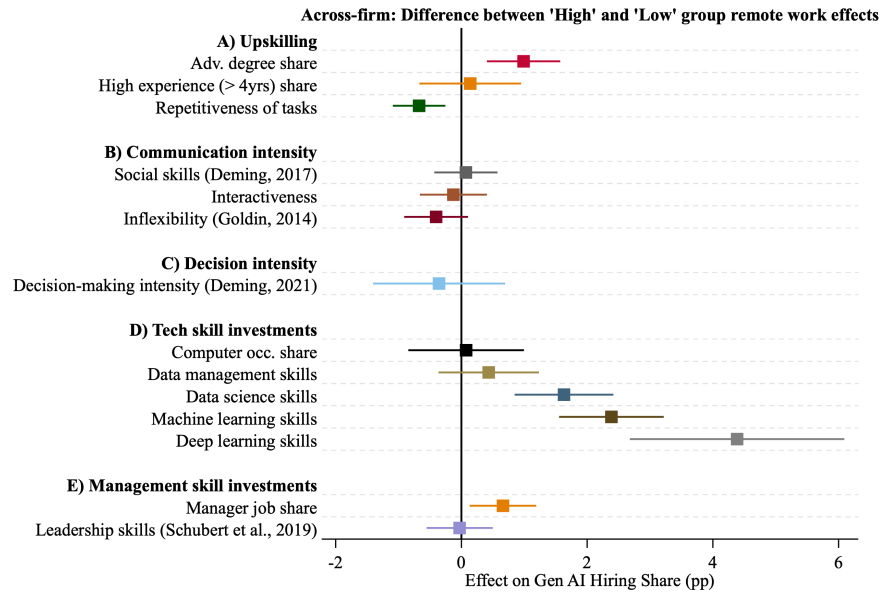


Figure 6: Remote work effects on Gen. AI adoption: heterogeneity by firm characteristics

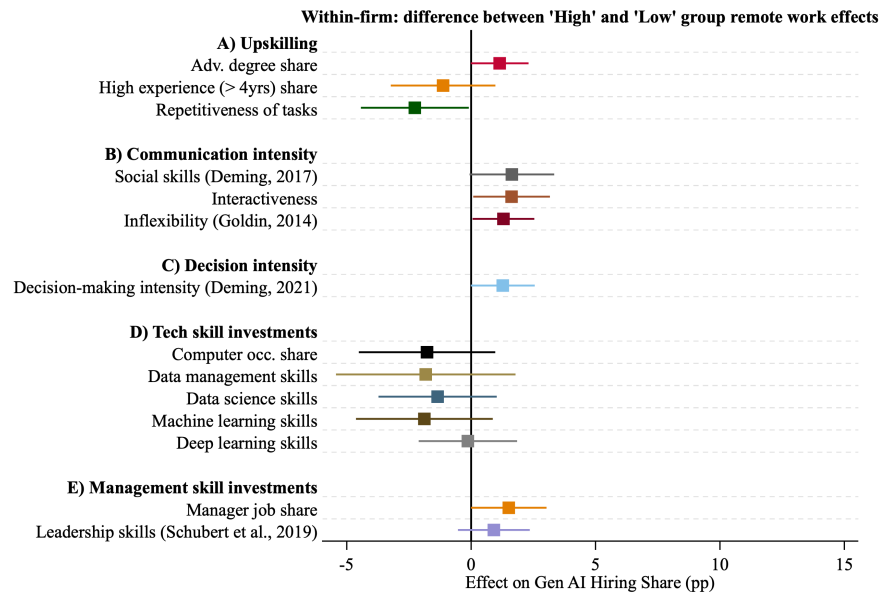
This figure shows coefficients estimated using IV for the effect of remote work prevalence in a firm's 2021/2022 (excl. Q4 2022) job postings on the prevalence of generative AI mentions in a firm's job postings in 2023/2024 (excl. Q4 2024) in a regression of the form:

$$100 \times \text{GenAIJobShare}('23-'24)_i = \alpha_{ind} + \beta \text{RWS}('21-'22)_i + \gamma \text{RWS}('21-'22)_i \times \mathbb{1}[\text{High}] + \text{Controls}_i + \varepsilon_i,$$

where *RWS* is the RemoteWorkShare and the dependent variable has been scaled by 100 for better readability. So, a coefficient of 10 indicates that a 10 pp change in remote work causes a 1pp change in generative AI adoption. $\mathbb{1}[\text{High}]$ indicates whether a firm is above average in the characteristic. The characteristics of jobs are measured in 2022 in Lightcast data, or are time-invariant characteristics based on 2018 O*Net data, and are computed from 2022 averages over a firm's hiring composition. Panel A shows the difference in effects between high and low group firms (corresponding to allowing for heterogeneity in the effect in column 2 of Table II), and panel B shows the difference in the within-firm effects across occupations between high and low group firms (corresponding to allowing for heterogeneity in the effect in column 4 of Table II). The control variables are the same as in Table II, and so are the instruments, except for adding an interaction between the $\mathbb{1}[\text{High}]$ indicator and the respective instrument. The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the firm level (panel A) or double-clustered at the occupation and firm level (panel B).



(A) Across-firm estimate of difference in effects (γ)



(B) Within-firm estimate of difference in effects (γ)

Figure 7:
Gen. AI exposure effects on remote work by firm characteristics

This figure shows the synthetic diff-in-diff effect estimates of the difference between the treated and control groups in remote work shares after the release of ChatGPT for different subsets of firms based their hiring characteristics. The treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). The subgroups are defined as firms that are above-median (square red markers) or below-median (grey triangles) in terms of each characteristic of firm hiring in 2022. The 95% confidence intervals shown are based on block bootstrap standard errors computed from 50 resamples, using the procedure detailed in Clarke et al. (2023).

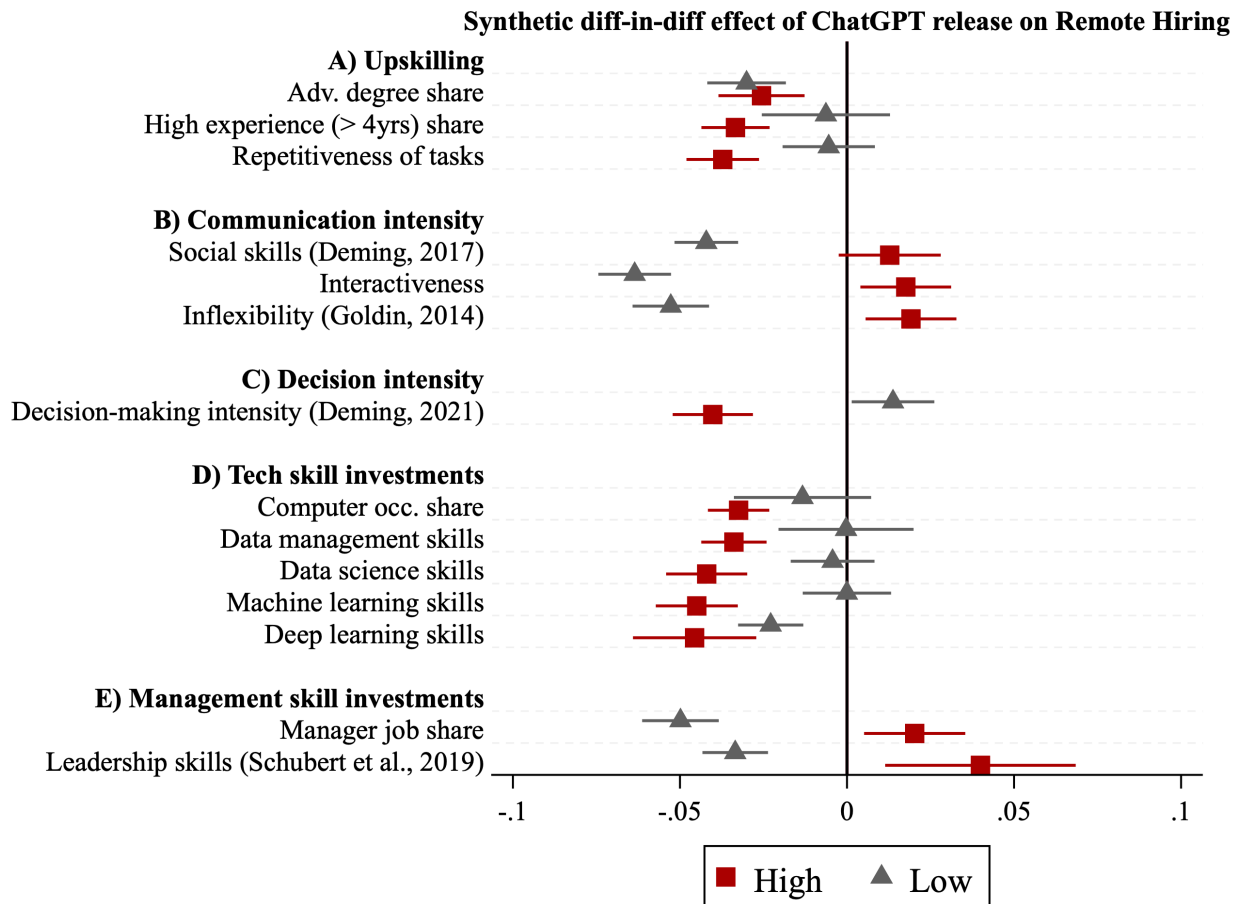


Table I:
Summary Statistics

This table reports summary statistics of key variables at the firm-level in Panel A and at the occupation-by-firm level in Panel B. See Section I for data sources and variable definitions.

	Mean	SD	10th	50th	90th	99th	99.9th	Obs.
<i>Panel A: Firm-level measures</i>								
Remote Work Share '19 (%)	1.295	7.157	0.00	0.00	1.18	33.33	100.00	87,032
Remote Work Share '21-'22 (%)	6.001	14.178	0.00	0.00	18.87	74.67	100.00	87,032
Fully Remote Work Share '21-'22 (%)	4.533	12.275	0.00	0.00	12.66	67.46	100.00	87,032
Hybrid Remote Work Share '21-'22 (%)	1.468	5.991	0.00	0.00	2.78	30.77	71.43	87,032
Generative AI Share '23-'24 (%)	0.090	1.090	0.00	0.00	0.00	2.37	11.84	87,032
Total Job Postings '23-'24	194.482	1261.407	11.00	30.00	246.00	3088.00	17310.00	87,032
Firm Teleworkability '23-'24	0.500	0.293	0.11	0.48	0.92	1.00	1.00	87,032
Firm Teleworkability '19	0.526	0.328	0.00	0.52	1.00	1.00	1.00	87,032
Firm Generative AI Exposure '23-'24	0.336	0.126	0.18	0.33	0.51	0.60	0.72	87,032
Firm Generative AI Exposure '19	0.349	0.150	0.16	0.35	0.54	0.73	0.87	87,032
1[Tech sector]	0.163	0.369	0.00	0.00	1.00	1.00	1.00	87,032
MSA Teleworkability Exposure '19	0.510	0.080	0.41	0.51	0.61	0.69	0.72	87,032
MSA Remote Share Exposure '19	2.381	0.917	1.52	2.31	3.22	5.42	13.80	87,032
MSA Commute Time Exposure '19	29.345	4.650	23.68	29.12	34.61	39.68	39.68	87,032
1[Return to Office]	0.009	0.093	0.00	0.00	0.00	0.00	1.00	87,032
College Degree Required Share '23-'24	0.264	0.267	0.00	0.18	0.69	0.95	1.00	87,032
Adv. Degree Required Share '23-'24	0.030	0.082	0.00	0.00	0.09	0.41	0.78	87,032
Data Mgmt. Skill Share '19	2.536	7.660	0.00	0.00	7.97	33.33	100.00	87,032
Machine Learning Skill Share '19	0.325	2.949	0.00	0.00	0.00	7.02	50.00	87,032
AI Skill Share '19	0.183	2.428	0.00	0.00	0.00	3.25	33.36	87,032
Deep Learning Skill Share '19	0.046	1.085	0.00	0.00	0.00	0.60	7.69	87,032
<i>Panel B: Occupation-by-firm-level measures</i>								
Remote Work Share '21-'22 (%)	6.156	18.974	0.00	0.00	19.35	100.00	100.00	1,495,607
Fully Remote Work Share '21-'22 (%)	4.979	17.041	0.00	0.00	11.54	100.00	100.00	1,495,607
Hybrid Remote Work Share '21-'22 (%)	1.178	8.234	0.00	0.00	0.00	40.00	100.00	1,495,607
Generative AI Share '23-'24 (%)	0.125	2.682	0.00	0.00	0.00	0.00	40.00	1,495,607
Total Job Postings '23-'24	10.872	105.311	1.00	2.00	15.00	132.26	873.00	1,495,607
1[Tech sector]	0.139	0.346	0.00	0.00	1.00	1.00	1.00	1,495,607
College Degree Required Share '23-'24	0.342	0.430	0.00	0.00	1.00	1.00	1.00	1,495,607
Adv. Degree Required Share '23-'24	0.039	0.171	0.00	0.00	0.00	1.00	1.00	1,495,607

Table II:
Remote work effects on Gen. AI Adoption

This table shows estimates of the remote share effect at the firm- and occupation-by-firm level in specifications of the form

$$100 \times \text{GenAIJobShare}('23-'34)_i = \beta \text{RemoteWorkShare}('21-'22)_i + \alpha_{\text{firm}} + \alpha_{\text{occ}} + \text{Controls}_i + \varepsilon_i$$

for the generative AI share for the YTD Sep. 2024 period and the remote share during 2021-2022 (excl. Q4 2022). The instrument consists of the interaction between firm-level exposure to MSA teleworkability through its hiring labor markets (measured in 2019) and occupation-level remote work adoption as of 2021-2022. The definition of baseline characteristics included as control can be found in Section III.B. T-test statistics based on heteroskedasticity-robust standard errors clustered at the firm level (columns 1 and 2) or double-clustered at the company and 6-digit occupation level (columns 3 and 4) in parentheses: * p<0.10, ** p<0.05, *** p<0.01.

<i>Dependent variable:</i>	100 × Generative AI Job Share (%)			
<i>Unit:</i>	Firm		Firm × Occupation	
<i>Estimation:</i>	OLS (1)	IV (2)	OLS (3)	IV (4)
Remote Job Share (%)	0.35*** (4.92)	4.06*** (7.35)	0.05* (1.65)	10.92** (2.30)
Observations	87,032	87,032	1,495,607	1,345,725
1st-stage KP F-stat.		417		29
Firm baseline characteristics	X	X		
2-dig. Industry FEs	X	X		
Firm × Occ. adv. educ. requirements			X	X
Occupation FEs			X	X
Firm FEs			X	X
<i>Instrument</i>	MSA Telework. Exposure × Firm Teleworkability		MSA Telework. Exposure × Firm Teleworkability × Occ. Hybrid Share	

Table III:
Robustness checks: IV remote work effects on Gen. AI Adoption

This table shows estimation results for the following regression specifications:

The specification in columns 1 and 3 is identical to the one in column 2 of Table II and the specifications in columns 2 and 4 is identical to that in column 4 of Table II, but with different measures of remote work: the independent variable in columns 1 and 2 are fully remote jobs, and in columns 3 and 4 it measures only hybrid jobs. Columns 5 and 6 interact both the independent variable and the instrument with an indicator for whether the firm is in potentially in the tech sector (NAICS 51 and 54). Columns 7 and 8 use an alternative instrument that is based on a firm's labor market exposure to different MSA commuting times, rather than teleworkability. Columns 9, 10, and 11 use data that is disaggregated to the MSA level and include MSA fixed effects. T-test statistics based on heteroskedasticity-robust standard errors in parentheses that are: clustered at the firm level (columns 1, 3, 5, 7); or double-clustered at the company and 6-digit occupation level (column 2, 4, 6, 8); or double-clustered at the company and MSA level (column 9, 10); or triple-clustered at the company and 6-digit occupation and MSA level (column 11): * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Dependent variable:													
Specification:	100 × Generative AI Job Share (%)												
	Fully remote only			Hybrid only		Tech sector			Alternative IV			MSA FEs	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)		
Remote Job Share (%)	5.29*** (7.26)	31.37*** (2.18)	17.50*** (6.07)	47.01*** (2.13)	3.53*** (5.74)	10.41*** (2.27)	5.28*** (4.37)	12.18*** (2.31)	5.00*** (6.38)	8.31*** (3.10)	24.39*** (2.07)		
Remote Job Share (%) x 1[Tech Sector]					0.97*** (3.18)	0.47 (0.53)							
Observations	87,032	1,345,725	87,032	1,345,725	87,032	1,345,725	87,032	1,345,725	627,055	623,808	2,519,136		
1st-stage KP F-stat.	315	10	99	19	172	16	78	24	113	44	6		
Baseline firm characteristics	X		X		X		X		X				
Firm × Occ. adv. educ. requirements		X		X		X		X		X			
Firm × MSA adv. educ. requirements													
Firm × Occ. × MSA adv. educ. requirements													
2-dig. Industry FEs	X		X		X		X		X		X		
Occupation FEs		X		X		X		X			X		
Firm FEs		X		X		X		X		X	X		
MSA FEs									X	X	X		
Unit:	Firm	Firm × Occ	Firm	Firm × Occ	Firm	Firm × Occ	Firm	Firm × Occ	Firm × MSA	Firm × MSA	Firm × Occ × MSA		
Instrument	MSA Tele. Exp. × Firm Tele.	MSA Tele. Exp. × Firm Tele. × Occ. Fully Remote Share	MSA Tele. Exp. × Firm Tele.	MSA Tele. Exp. × Firm Tele. × Occ. Hybrid Share	MSA Tele. Exp. × Firm Tele.	MSA Tele. Exp. × Firm Tele.	MSA Tele. Exp. × Firm Tele. × Occ. Hybrid Share	MSA Commute Time Exp. × Firm Tele. × Occ. Hybrid Share	MSA Tele. × Firm Tele.	MSA Tele. × Firm Tele.	MSA Tele. × Firm Tele.	MSA Tele. × Firm Tele.	

Table IV:
Synthetic diff-in-diff event study: effects of GenAI exposure

This table shows estimates of the effect of being in the high generative AI exposure group post-ChatGPT release, estimated using the synthetic diff-in-diff estimator in equation 2. The treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). The dependent variable is the share of all jobs that are remote in column 1, and the absolute number of job postings that are remote, or of all types, in columns 2 and 3. T-test statistics based on standard errors from a block bootstrap procedure with 50 resamples at the company level in parentheses: * p<0.10, ** p<0.05, *** p<0.01.

<i>Dependent variable:</i>	<i>% of Jobs</i>	<i>Total Jobs</i>	
<i>Job category:</i>	Remote	Remote	All
	(1)	(2)	(3)
<i>Panel A: All firms</i>			
$\mathbb{1}[\text{Post-ChatGPT}] \times \mathbb{1}[\text{High Gen. AI Exposure}]$	-0.027*** (-6.02)	-11.237** (-2.55)	-29.727*** (-2.81)
Observations	222,555	222,555	222,555
<i>Panel B: Tech sector (NAICS 51 & 54)</i>			
$\mathbb{1}[\text{Post-ChatGPT}] \times \mathbb{1}[\text{High Gen. AI Exposure}]$	-0.109*** (-8.87)	-13.230 (-1.14)	5.751 (0.17)
Observations	32,835	32,835	32,835
<i>Panel C: Non-Tech sectors</i>			
$\mathbb{1}[\text{Post-ChatGPT}] \times \mathbb{1}[\text{High Gen. AI Exposure}]$	0.014*** (3.17)	-10.110 (-1.56)	-53.447*** (-3.25)
Observations	189,720	189,720	189,720
<i>Estimation:</i>	SDID	SDID	SDID
Firm FEs	X	X	X
Time FEs	X	X	X

Table V:
Synthetic diff-in-diff event study: robustness checks

This table shows additional specifications that use synthetic difference-in-differences design to estimate the effect of being in the high generative AI exposure group post-ChatGPT release, as shown in equation 2. The dependent variable is the share of all jobs that are remote, and the specifications are the same as in column (1) of Table IV, except for the following variations: Column (1) and (2) change the cutoff that defines “high” generative AI exposure to be the 95th or 85th percentile of firms. Column (3) residualizes the dependent variable with regard to 2-digit-industry-by-year fixed effects, and column (4) defines the full time period for the event study to be the 15 quarters Q4 2013-Q2 2017, and the post period as the period Q3 2015-Q2 2017. T-test statistics based on standard errors from a block bootstrap procedure with 50 resamples at the company level in parentheses: * p<0.10, ** p<0.05, *** p<0.01.

<i>Dependent variable:</i>	Remote Share of All Jobs			
<i>Specification:</i>	95th Pctl Exp. Cutoff (1)	85th Pctl Exp. Cutoff (2)	2-dig. Ind× Year FEs (3)	Placebo× Event (4)
<i>Panel A: All firms</i>				
1[Post]× 1[High Exp.]	-0.064*** (-10.50)	-0.007* (-1.91)	-0.023*** (-5.00)	0.002 (1.25)
Observations	222,555	222,555	208,410	76,125
<i>Panel B: Tech sector (NAICS 51 & 54)</i>				
1[Post]× 1[High Exp.]	-0.098*** (-5.52)	-0.104*** (-11.63)	-0.108*** (-7.34)	-0.002 (-0.38)
Observations	32,835	32,835	32,835	12,630
<i>Panel C: Non-Tech sectors</i>				
1[Post]× 1[High Exp.]	0.000 (0.06)	0.012*** (3.27)	0.007 (1.39)	0.001 (0.95)
Observations	189,720	189,720	175,575	63,495
<i>Estimation:</i>	SDID	SDID	SDID	SDID
Firm FEs	X	X	X	X
Time FEs	X	X	X	X

Table VI:
Change in firm-level job skill demand after GenAI adoption

This table shows estimates of the effect of being in the high generative AI exposure group post-ChatGPT release, estimated using the synthetic diff-in-diff estimator in equation 2. The treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). The dependent variables represent shares of different job categories among all non-remote workers (columns 1-5) and remote workers (columns 6-10). Panel A shows results among all firms, panel B limits the sample to the tech sector, and panel C only includes non-tech firms. Each panel only including only firms that have non-remote or remote job postings, respectively, in all quarters included in the sample (Q1 2021-Q3 2024). T-test statistics based on standard errors from a block bootstrap procedure with 50 resamples at the company level in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

<i>Dep. var. jobs:</i>	Share of Non-remote jobs					Share of Remote jobs				
	High Experience (1)	Advanced degree (2)	Data mgmt. (3)	Decision making (4)	Communication (5)	High Experience (6)	Advanced degree (7)	Data mgmt. (8)	Decision making (9)	Communication (10)
<u>Panel A: All firms</u>										
$\mathbb{1}[\text{Post}] \times \mathbb{1}[\text{High Exp.}]$	0.016*** (5.80)	0.000 (0.52)	0.002** (2.07)	0.014*** (4.31)	0.007** (2.06)	0.012** (2.22)	0.000 (0.03)	0.001 (0.31)	0.028*** (4.76)	0.027*** (4.71)
Observations	220,410	220,410	220,410	220,410	220,410	41,340	41,340	41,340	41,340	41,340
<u>Panel B: Tech sector</u>										
$\mathbb{1}[\text{Post}] \times \mathbb{1}[\text{High Exp.}]$	-0.009 (-1.45)	0.001 (0.49)	0.001 (0.31)	0.030*** (3.32)	0.011 (1.52)	-0.019 (-1.63)	-0.002 (-0.32)	-0.006 (-0.77)	0.047*** (3.70)	0.030*** (2.66)
Observations	32,010	32,010	32,010	32,010	32,010	12,315	12,315	12,315	12,315	12,315
<u>Panel C: Non-tech sector</u>										
$\mathbb{1}[\text{Post}] \times \mathbb{1}[\text{High Exp.}]$	0.011*** (5.63)	-0.002** (-2.41)	-0.002** (-2.00)	-0.006* (-1.91)	-0.003 (-0.96)	0.008 (1.08)	0.004* (1.79)	0.001 (0.23)	-0.005 (-0.69)	0.004 (0.59)
Observations	188,400	188,400	188,400	188,400	188,400	29,025	29,025	29,025	29,025	29,025
<i>Estimation:</i>	SDID	SDID	SDID	SDID	SDID	SDID	SDID	SDID	SDID	SDID
Firm FEs	X	X	X	X	X	X	X	X	X	X
Time FEs	X	X	X	X	X	X	X	X	X	X

Table VII:
Return-to-Office firms have higher remote work effects on Gen. AI Adoption

This table shows estimation results for the following regression specifications: The specification in column 1 is identical to the one in column 2 of Table II and the specification in column 2 is identical to that in column 4 of Table II, except for the following differences: the endogenous remote share is interacted with an indicator for a company that has a return-to-office (RTO) policy as of March 2025, and so is the instrument for the endogenous variable; an indicator for a firm having an RTO policy is added as an additional control variable where this is not collinear with fixed effects. T-test statistics based on heteroskedasticity-robust standard errors clustered at the firm level (column 1) or double-clustered at the company and 6-digit occupation level (column 2) in parentheses: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

<i>Dependent variable:</i>	100 × Generative AI Job Share (%)	
<i>Estimation:</i>	IV (1)	IV (2)
Remote Job Share (%)	3.99*** (7.23)	10.83** (2.27)
Remote Job Share (%) x $\mathbb{1}[\text{Return-to-Office}]$	6.08*** (4.10)	1.29 (1.07)
Observations	87,032	1,345,725
1st-stage KP F-stat.	208	14
“Firm has RTO Policy” Indicator	X	
Baseline firm characteristics	X	
2-dig. Industry FEs	X	
Firm × Occ. adv. educ. requirements		X
Occupation FEs		X
Firm FEs		X

A Appendix Figures & Tables

Figure A.1:
Remote work and generative AI adoption by industry sector

This figure plots the share of job postings in the YTD Sep. 2024 period that are for jobs that mention generative AI relative to those that are for remote jobs. The job postings data are from Lightcast and are aggregated into 2-digit NAICS industry sectors. The red line indicates a linear best fit.

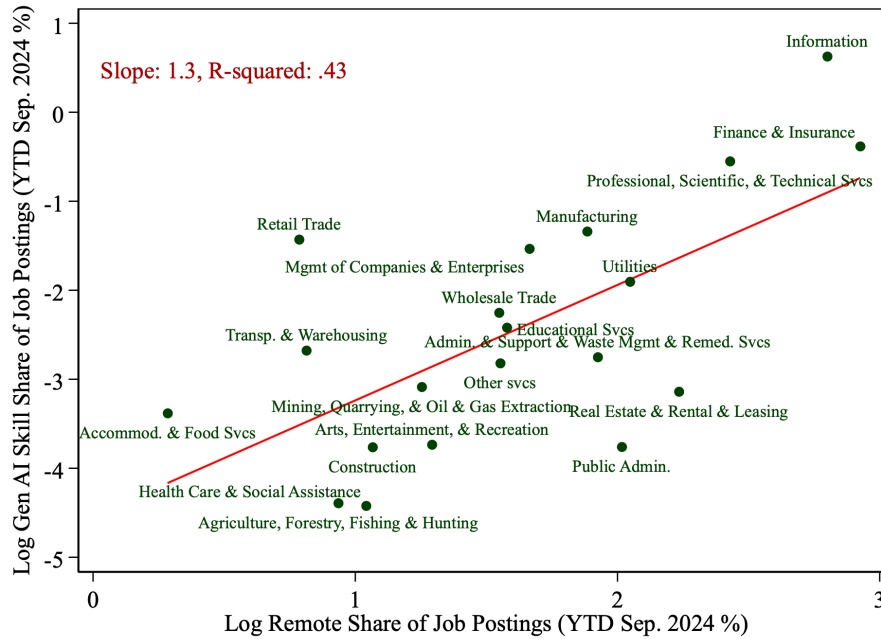


Figure A.2:
Remote work and generative AI at the firm level

This figure plots the share of job postings in the YTD Sep. 2024 period that are for jobs that mention generative AI relative to those that are for remote jobs. The job postings data are from Lightcast and are aggregated by company. The red line indicates a linear best fit. The analysis only includes companies that have at least 10 job postings in the sample period shown and have non-zero job postings in all quarters from Q1 2021 to Q3 2024.

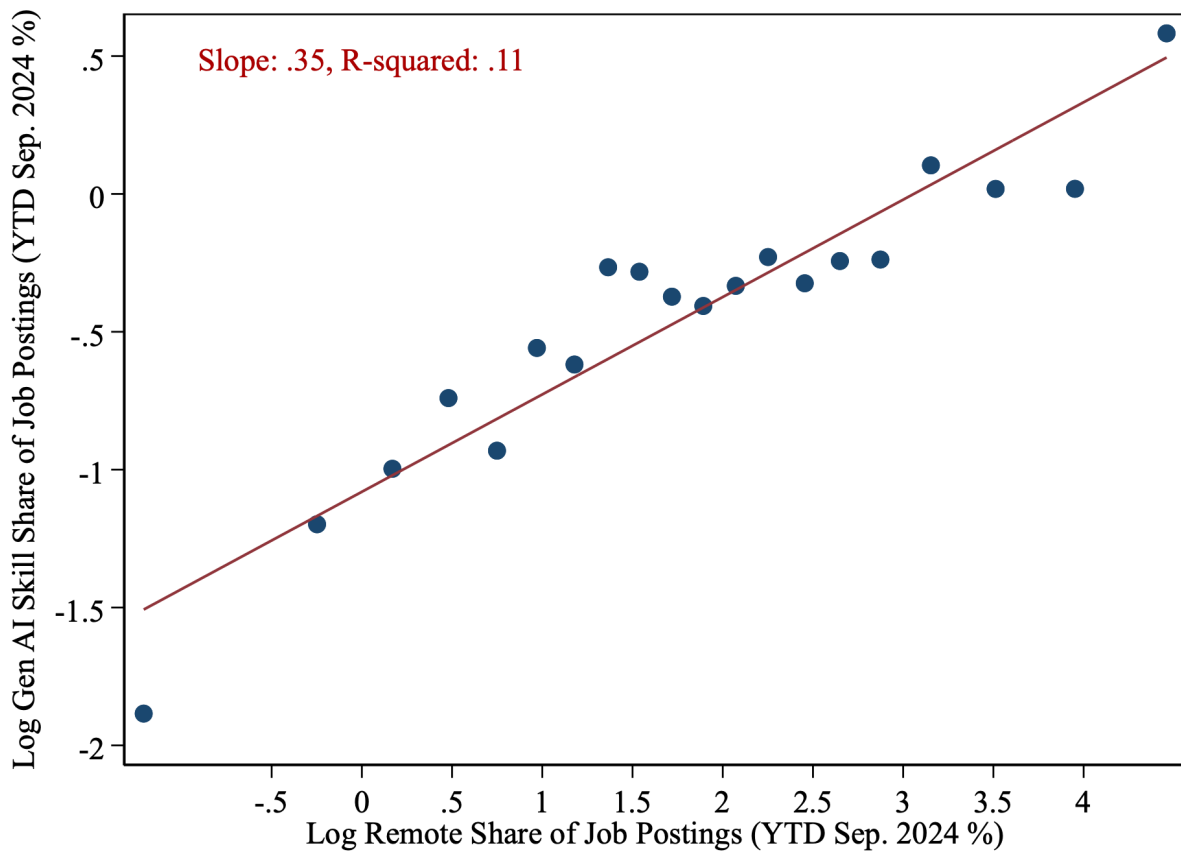
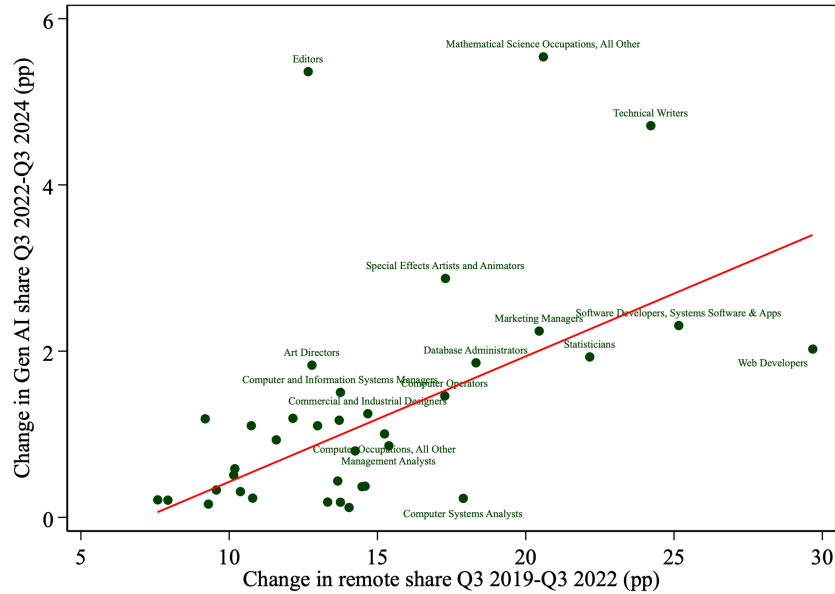
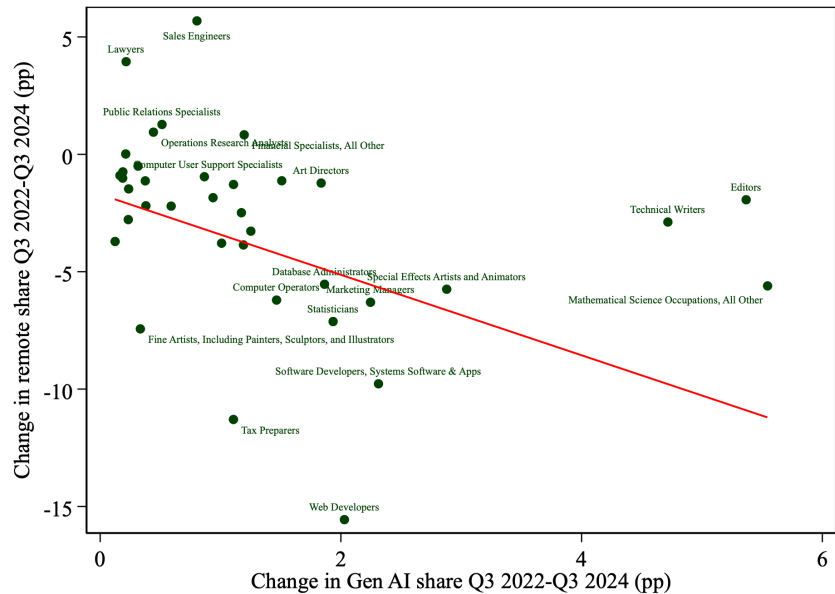


Figure A.3:
Pandemic remote hiring and generative AI adoption

These figures focus on the top quartile of remote work adopting occupations (based on 2021/2022 job postings). Panel A shows the change in hiring for generative AI skills after the release of ChatGPT (Q3 2022-Q3 2024) as a function of remote work changes during the pandemic (Q3 2019-Q3 2022). Panel B shows remote work changes after the ChatGPT release as a function of generative AI adoption over the same period. Both graphs only include occupations with > 1K job postings in Q3 2022 with some Q3 2024 Gen AI adoption (> 0.1% of jobs). Two outlier occupations, 'Computer Programmers' and 'Writers & Authors' are not included for better visibility, but follow similar patterns. The data are Lightcast job postings where “Gen AI share” is the share of jobs mentioning Gen. AI-related keywords.



(A) Δ Remote Work ('19-'22) and Gen. AI adoption ('22-'24)



(B) Gen. AI adoption ('22-'24) and Δ Remote Work ('22-'24)

Figure A.4:
Remote Work Suitability and Generative AI Exposure by Occupation

This figure shows the relation between Generative AI exposure and remote work suitability. Generative AI exposure and teleworkability are measured at the SOC 2010 6-digit occupation level based on data from Eisfeldt et al. (2023) and Dingel and Neiman (2020), and aggregated across detailed occupations based on employment weights as of 2022 from the Occupational Employment Statistics. The line of best fit in red is estimated with employment weights.

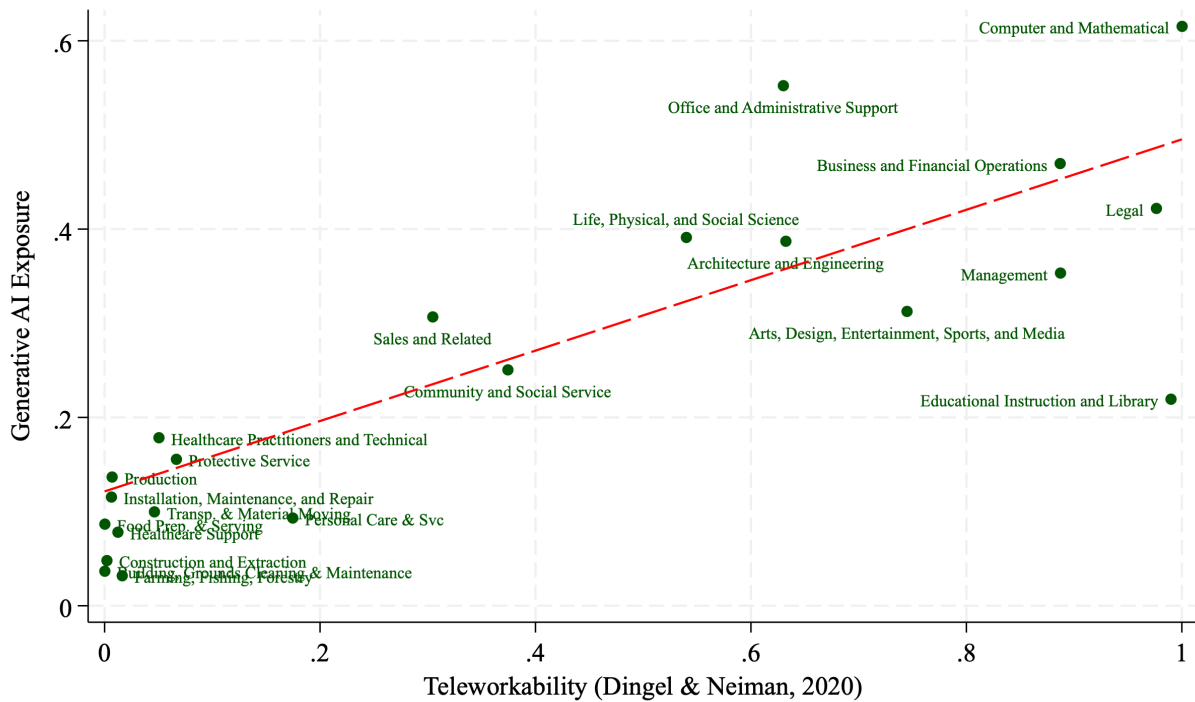


Figure A.5:
Generative AI exposure and generative AI adoption by firms

These graphs show the relationship between the share of job postings at the firm level that mention generative AI skills in the YTD Sep. 2024 period, and the Eisefeldt et al. (2023) measure of generative AI exposure of the firm's job postings. The vertical axis shows the share of generative AI skill mentions in all job postings. Firms are sorted into deciles by exposure and each bar shows the total job posting-weighted mean share of all posted jobs at firms in that decile of exposure that mention generative AI.

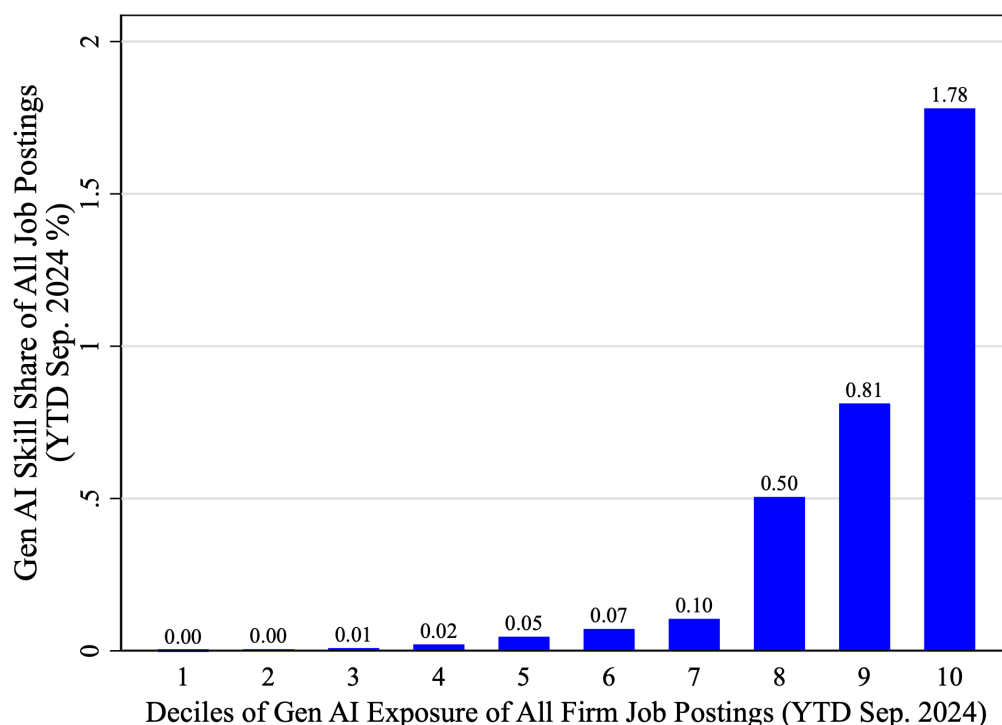


Figure A.6:
Remote work effects on Gen. AI adoption by industry sector

This figure shows coefficients estimated using IV for the effect of remote work prevalence in a firm's 2021/2022 (excl. Q4 2022) job postings on the prevalence of generative AI mentions in a firm's job postings in 2023/2024 (excl. Q4 2024). The specification is the same as in columns (2) of Table 1, but restricting the sample of firms to different industry sectors. Firms are categorized as follows into broad sectors (using 2-digit NAICS codes): Professional & Technical Services (NAICS 51 and 54); Healthcare & Social Assistance (NAICS 62); Construction & Real Estate (NAICS 22, 23, and 53); Manufacturing (NAICS 21, 31, 32, and 33); Business Support & Financial Services (NAICS 52, 55, and 56); Education & Other Services (NAICS 61 and 81); Trade & Transportation (NAICS 42, 44, 45, 48, and 49); Hospitality & Entertainment (NAICS 71 and 72). The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the firm level.

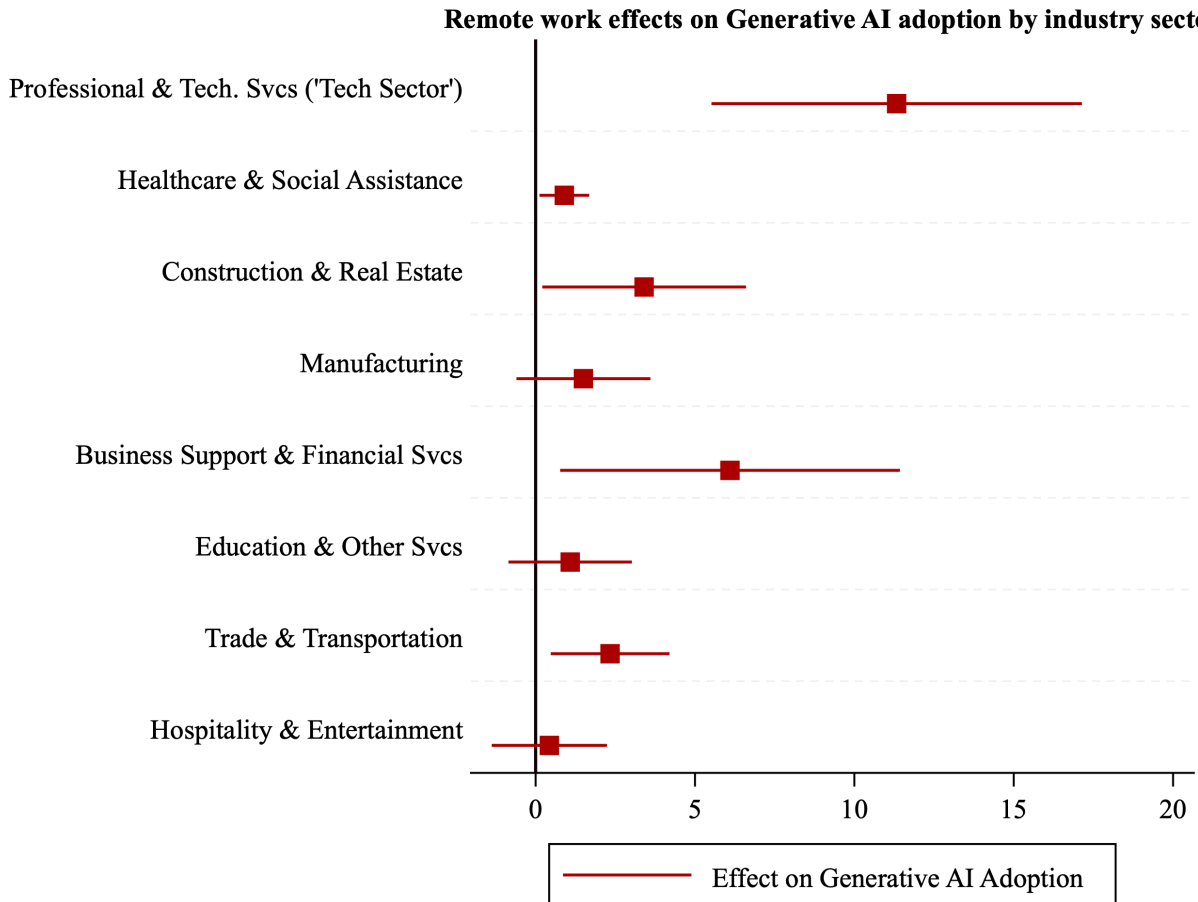


Figure A.7:
Remote work trends by firm-level generative AI exposure

This figure shows the share of job postings in each quarter that are for jobs that are for remote jobs. In each panel, the jobs are aggregated into job posting-weighted quartiles of the firms' Eisefeldt et al. (2023) generative AI exposure, based on job postings in 2021-2022 (excl. Q4 2022). The grey drop line indicates the period (Q4 2022) when ChatGPT was released.

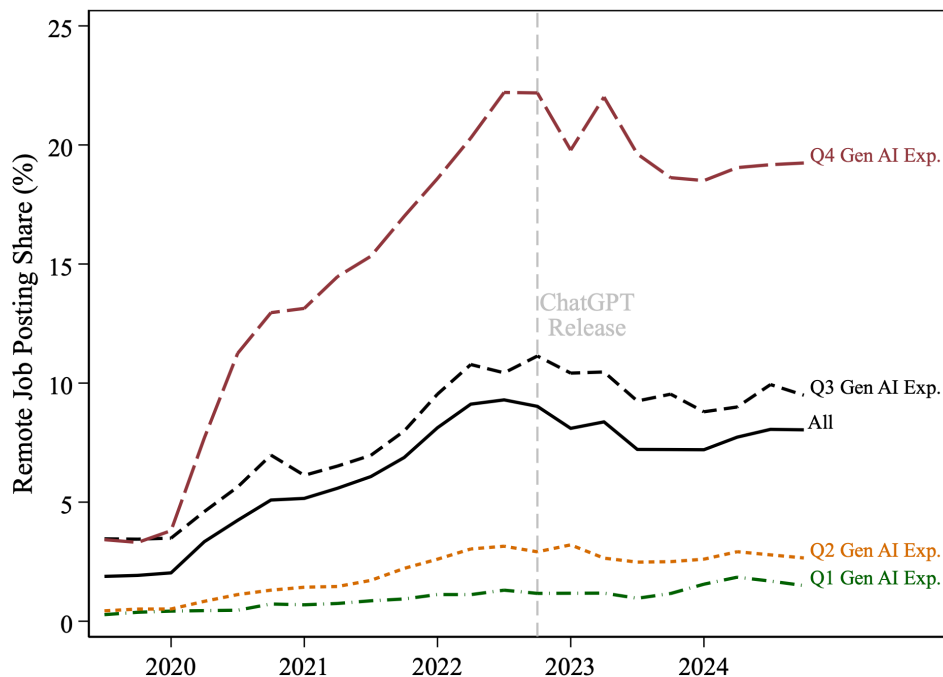


Figure A.8:
Firm characteristics and remote work adoption

This figure shows OLS estimates of coefficients for the effect of standardized firm characteristics in 2019 on the firm's remote work prevalence in 2021/2022 (excl. Q4 2022) job postings in a regression of the form

$$\text{RemoteWorkShare}('21-'22)_i = \alpha_{ind} + \beta \text{FirmCharacteristic}(2019)_i + \text{Controls}_i + \varepsilon_i,$$

where the controls in all regressions include the 2019 value of the dependent variable as a control variable, so the coefficients can be interpreted as the effect of a standardized difference in the characteristic on changes in the remote work share. The control variables also include NAICS 2-digit fixed effects, firm-level teleworkability in 2019 and 2021/2022; and the company's share of jobs requiring a college education in 2019. The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the NAICS 2-digit sector level.

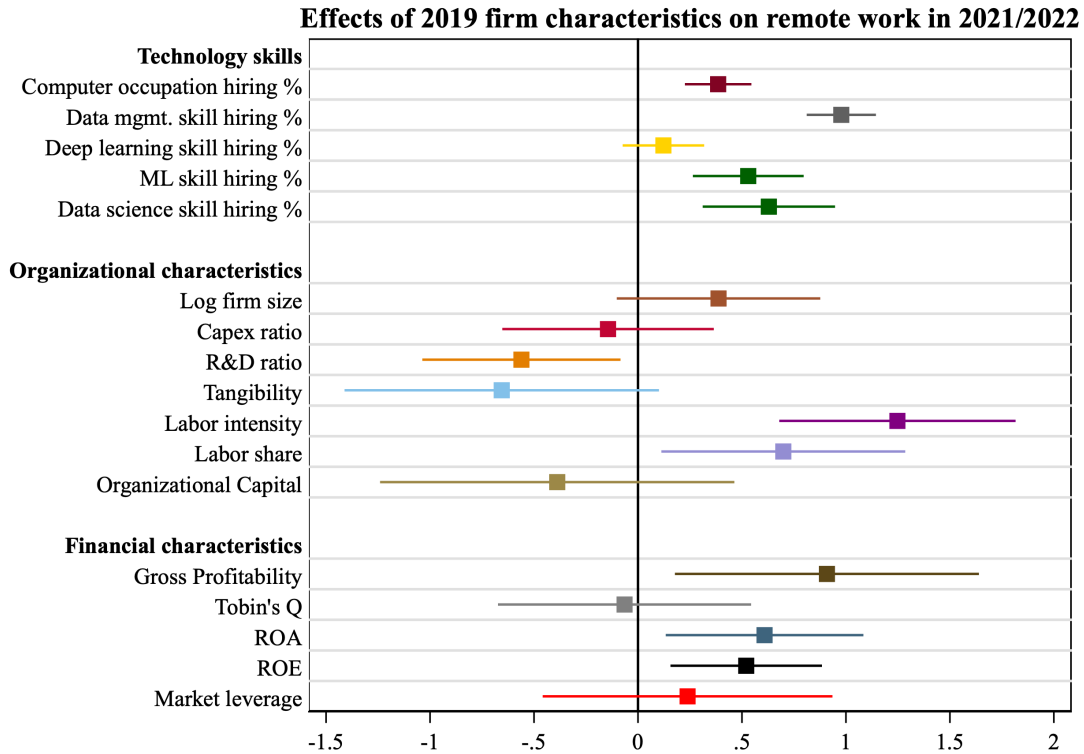


Figure A.9:
Across-firm Gen. AI exposure effects on Gen. AI adoption by firm characteristics

This figure shows the the cross-sectional estimates of the differential effect of generative AI exposure in 2021/2022 on generative AI adoption in 2023/24 based on the hiring characteristics of a firm in 2022. Each row in the figures shows coefficient estimates corresponding to a different specification of the form

$$100 \times \text{GenAIJobShare}('23-'24)_i = \alpha_{ind} + \beta \text{GenAIExp}('21-'22)_i + \gamma \text{GenAIExp}('21-'22)_i \times \mathbb{1}[\text{High}_i] + \text{Controls}_i + \varepsilon_i,$$

where i is an occupation-by-firm unit and the dependent variable has been scaled by 100 for better readability. So, a coefficient of 10 indicates that a 10 pp change in generative AI exposure causes a 1pp change in generative AI adoption. $\mathbb{1}[\text{High}_i]$ indicates whether an occupation-by-firm cell is above average in the characteristic in the sample. The characteristics are either measured as means of the hiring characteristics of 2021/2022 Lightcast data, if they vary at the occupation-by-firm level, or are time-invariant based on 2018 O*Net data if they are occupation-level characteristics, which is the case for: repetitiveness, social skills, interactivensess, inflexibility, decision-making intensity, decision-making without supervision, computer occupation, manager occupation, and leadership skills. The top panel shows estimates of $\hat{\gamma}$ that correspond to the difference in effects between the high and the low group, and the bottom panel shows the baseline effect $\hat{\beta}$ for the low group. The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the firm level.

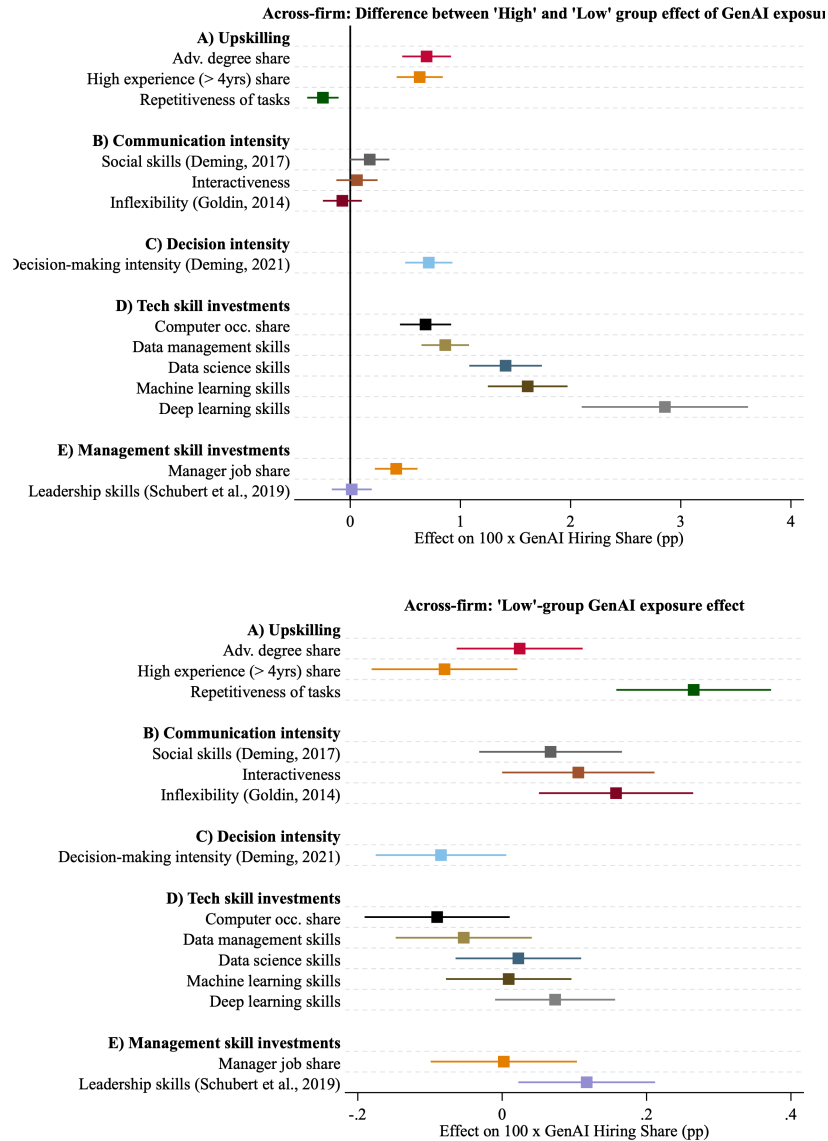


Figure A.10:
Within-firm Gen. AI exposure effects on Gen. AI adoption by occupation characteristics

This figure shows the the cross-sectional estimates of the differential effect of generative AI exposure in 2021/2022 on generative AI adoption in 2023/24 for different subsets of occupation-by-firm cells based on static characteristics of the occupation or the hiring characteristics of the occupation-by-firm in 2022. Each row in the figures shows coefficient estimates corresponding to a different specification of the form

$$100 \times \text{GenAIJobShare}('23-'24)_i = \alpha_{firm} + \beta \text{GenAIExp}('21-'22)_i + \gamma \text{GenAIExp}('21-'22)_i \times \mathbb{1}[\text{High}_i] + \text{Controls}_i + \varepsilon_i,$$

where i is an occupation-by-firm unit and the dependent variable has been scaled by 100 for better readability. So, a coefficient of 10 indicates that a 10 pp change in generative AI exposure causes a 1pp change in generative AI adoption. $\mathbb{1}[\text{High}_i]$ indicates whether an occupation-by-firm cell is above median in the characteristic in the sample. The characteristics are either measured as means of the hiring characteristics of 2021/2022 Lightcast data, if they vary at the occupation-by-firm level, or are time-invariant based on 2018 O*Net data if they are occupation-level characteristics, which is the case for: repetitiveness, social skills, interactiveness, inflexibility, decision-making intensity, decision-making without supervision, computer occupation, manager occupation, and leadership skills. The top panel shows estimates of $\hat{\gamma}$ that correspond to the difference in effects between the high and the low group, and the bottom panel shows the baseline effect $\hat{\beta}$ for the low group. All regressions control for firm fixed effects and the share of hiring at the occupation-by-firm level in 2023/24 that requires a college degree or an advanced degree. The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the firm level.

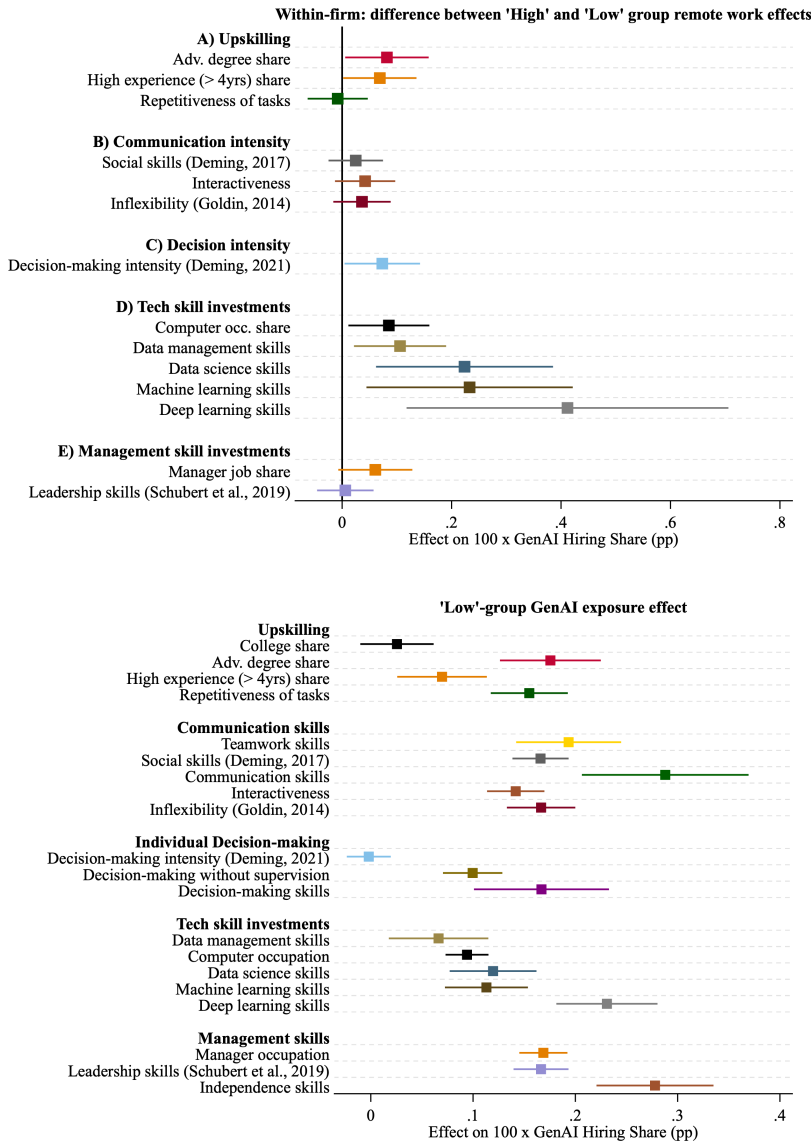
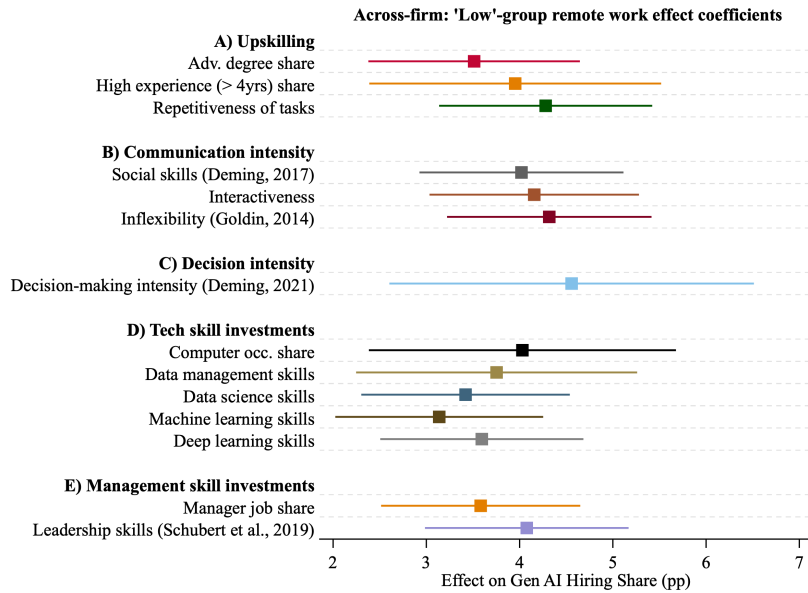


Figure A.11: Remote work effects on Gen. AI adoption: 'low'-group effects by firm characteristics

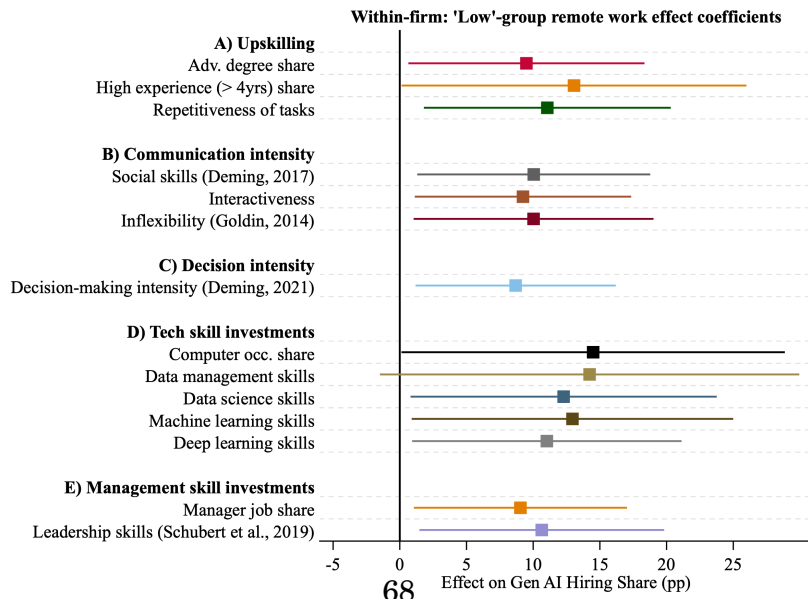
This figure shows coefficients estimated using IV for the effect of remote work prevalence in a firm's 2021/2022 (excl. Q4 2022) job postings on the prevalence of generative AI mentions in a firm's job postings in 2023/2024 (excl. Q4 2024) in a regression of the form:

$$100 \times \text{GenAIJobShare}('23-'24)_i = \alpha_{ind} + \beta \text{RWS}('21-'22)_i + \gamma \text{RWS}('21-'22)_i \times \mathbb{1}[\text{High}] + \text{Controls}_i + \varepsilon_i,$$

where *RWS* is the RemoteWorkShare and the dependent variable has been scaled by 100 for better readability. So, a coefficient of 10 indicates that a 10 pp change in remote work causes a 1pp change in generative AI adoption. $\mathbb{1}[\text{High}]$ indicates whether a firm is above average in the characteristic. The characteristics of jobs are measured in 2022 in Lightcast data, or are time-invariant characteristics based on 2018 O*Net data, and are computed from 2022 averages over a firm's hiring composition. Panel A shows the 'low'-group effects β across firms (corresponding to allowing for heterogeneity in the effect in column 2 of Table II), and panel B shows the within-firm effects across occupations for the 'low'-group firms (corresponding to allowing for heterogeneity in the effect in column 4 of Table II). The control variables are the same as in Table II, and so are the instruments, except for adding an interaction between the $\mathbb{1}[\text{High}]$ indicator and the respective instrument. The 95% confidence intervals shown are based on heteroskedasticity-robust standard errors clustered at the firm level (panel A) or double-clustered at the occupation and firm level (panel B).



(A) Across-firm estimate of 'low'-group effects (β)



(B) Within-firm estimate of 'low'-group effects (β)

Figure A.12:
Gen. AI exposure effects on remote work by industry sector

This figure shows the synthetic diff-in-diff effect estimates of the difference between the treated and control groups in remote work shares after the release of ChatGPT for different subsets of firms based on their 2-digit NAICS industry. After restricting the sample to the respective sector's firms, the treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). Firms are categorized as follows into broad sectors (using 2-digit NAICS codes): Professional & Technical Services (NAICS 51 and 54); Healthcare & Social Assistance (NAICS 62); Construction & Real Estate (NAICS 22, 23, and 53); Manufacturing (NAICS 21, 31, 32, and 33); Business Support & Financial Services (NAICS 52, 55, and 56); Education & Other Services (NAICS 61 and 81); Trade & Transportation (NAICS 42, 44, 45, 48, and 49); Hospitality & Entertainment (NAICS 71 and 72). The 95% confidence intervals shown are based on block bootstrap standard errors computed from 50 resamples, using the procedure detailed in Clarke et al. (2023).

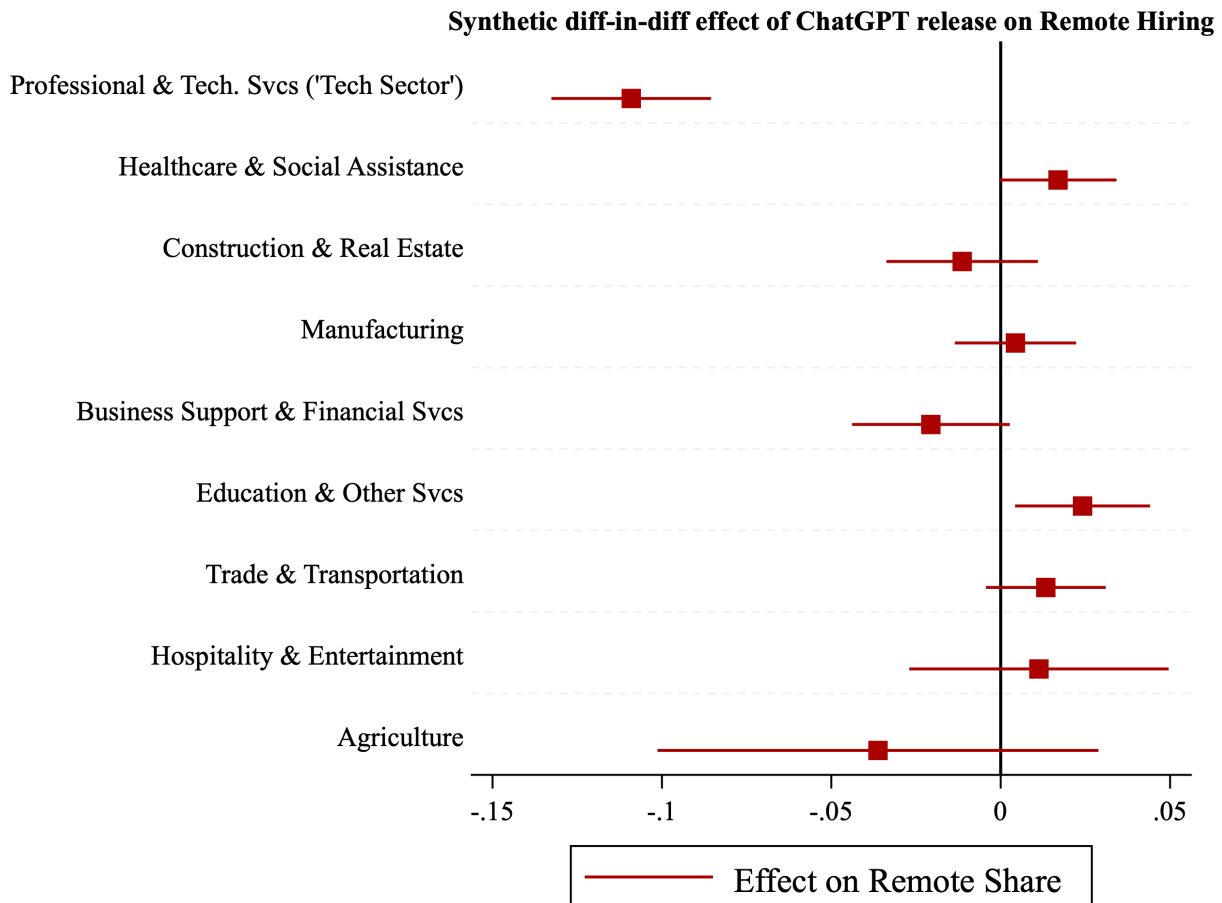


Figure A.13:
Gen. AI exposure effects on remote work by firm characteristics: Non-Tech Sectors

This figure shows the synthetic diff-in-diff effect estimates of the difference between the treated and control groups in remote work shares after the release of ChatGPT for different subsets of firms based their hiring characteristics. The treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). The subgroups are defined as firms that are above-median (square red markers) or below-median (grey triangles) in terms of each characteristic of firm hiring in 2022. The 95% confidence intervals shown are based on block bootstrap standard errors computed from 50 resamples, using the procedure detailed in Clarke et al. (2023).

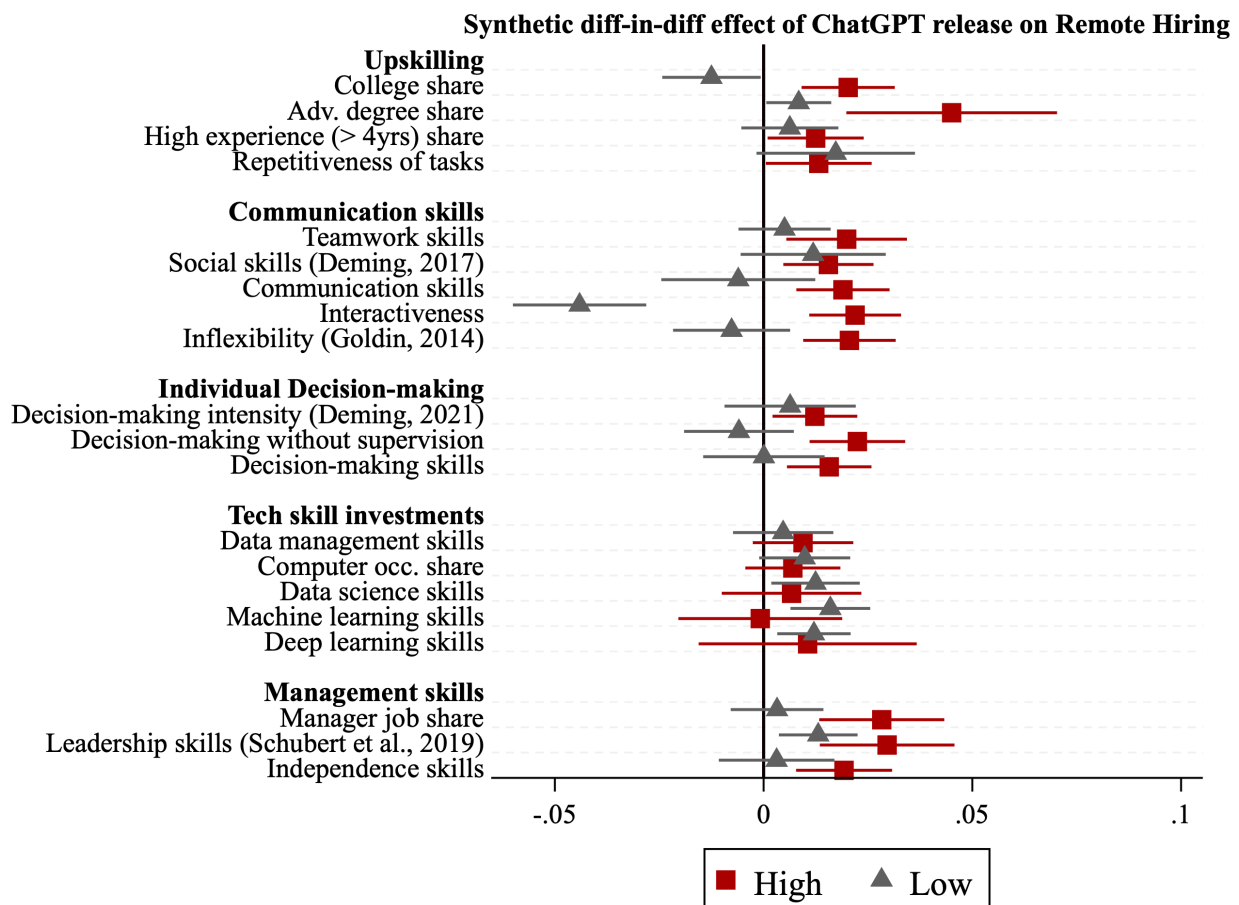


Figure A.14:

Gen. AI exposure effects on remote work by firm characteristics: Tech Sector

This figure shows the the synthetic diff-in-diff effect estimates of the difference between the treated and control groups in remote work shares after the release of ChatGPT for different subsets of firms based their hiring characteristics. The treatment group is defined as firms that are in the top decile of generative AI exposure based on the composition of their 2021/2022 job postings (excl. Q4 2022). The control group is determined by weighting firms outside the top decile to find the best fit to the pre-period trends in the treatment group, as described in Arkhangelsky et al. (2021). The subgroups are defined as firms that are above-median (square red markers) or below-median (grey triangles) in terms of each characteristic of firm hiring in 2022. The 95% confidence intervals shown are based on block bootstrap standard errors computed from 50 resamples, using the procedure detailed in Clarke et al. (2023).

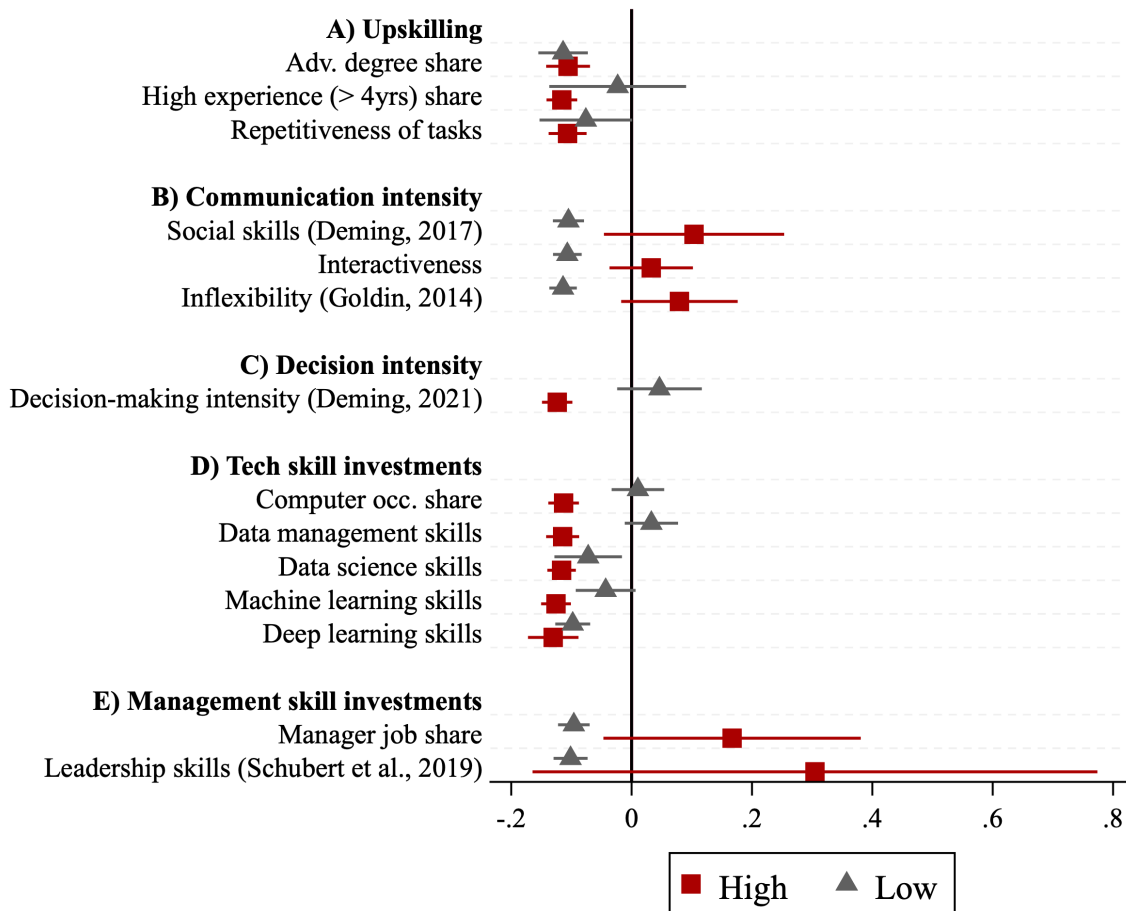


Table A.1:
Remote work and generative AI prevalence by occupation.

This table shows the share of all job postings in the YTD ending Sep. 2024 in each 6-digit SOC 2010 occupation that are for remote jobs (panel A), require generative AI-related skills (panel B), and are both remote and require generative AI skills (panel C). Each panel shows the top 20 occupations, ranked by the measure of interest, which have at least 5,000 job postings during the measurement period. The last column of the table shows the total job postings in the sample for that occupation.

(a) A: Remote shares

SOC	Occupation title	Remote	Jobs
15-2011	Actuaries	43	9,871
13-2053	Insurance Underwriters	31	19,385
27-3042	Technical Writers	30	19,584
41-9041	Telemarketers	30	6,026
21-1014	Mental Health Counselors	29	60,334
41-3041	Travel Agents	28	5,208
13-1031	Claims Adjusters, Examiners, and Investigators	28	58,753
43-9041	Insurance Claims and Policy Processing Clerks	26	20,677
13-2082	Tax Preparers	25	33,201
19-3094	Political Scientists	25	6,688
15-1134	Web Developers	23	69,144
15-1121	Computer Systems Analysts	23	78,807
27-3043	Writers and Authors	23	27,545
13-1075	Labor Relations Specialists	23	5,305
41-9031	Sales Engineers	23	13,462
23-1011	Lawyers	22	103,553
13-1111	Management Analysts	22	125,206
15-1122		21	36,023
15-1132	Software Developers, Applications	21	218,581
15-1133	Software Developers, Systems Software	21	218,581

(b) B: Generative AI shares

SOC	Occupation title	Gen. AI	Jobs
15-1131	Computer Programmers	8.5	19,768
27-3042	Technical Writers	6.1	19,584
27-3043	Writers and Authors	5.8	27,545
15-2099	Mathematical Science Occupations, All Other	4	187,460
27-3091	Interpreters and Translators	3	22,998
27-3041	Editors	1.9	16,550
11-2021	Marketing Managers	1.7	197,468
15-1133	Software Developers, Systems Software	1.7	218,581
15-1132	Software Developers, Applications	1.7	218,581
11-3021	Computer and Information Systems Managers	1.4	17,145
15-1134	Web Developers	1.4	69,144
15-1141	Database Administrators	1.2	133,219
27-1011	Art Directors	1.1	9,699
43-9011	Computer Operators	1.1	305,659
27-1014	Special Effects Artists and Animators	1.1	6,281
25-9031	Instructional Coordinators	.98	28,773
25-3099	Teachers and Instructors, All Other	.95	38,229
11-9041	Architectural and Engineering Managers	.95	65,851
27-1021	Commercial and Industrial Designers	.89	18,390
15-2041	Statisticians	.78	6,128

(c) C: Generative AI share in remote jobs

SOC	Occupation title	Gen. AI Remote	Jobs
27-3091	Interpreters and Translators	20	22,998
27-3042	Technical Writers	16	19,584
15-1131	Computer Programmers	9.6	19,768
27-3043	Writers and Authors	8.9	27,545
27-1014	Special Effects Artists and Animators	5.6	6,281
25-9021	Farm and Home Management Educators	4	20,921
15-2099	Mathematical Science Occupations, All Other	3.9	187,460
25-9031	Instructional Coordinators	3.1	28,773
11-9041	Architectural and Engineering Managers	3.1	65,851
25-2031	Sec. School Teachers (exc. Special/Tech. Ed.)	2.8	131,514
15-2041	Statisticians	2.5	6,128
15-1134	Web Developers	2.2	69,144
17-1011	Architects, Except Landscape and Naval	2	9,934
11-9032	Education Administrators, K-12	2	54,092
19-4021	Biological Technicians	1.9	7,372
17-2011	Aerospace Engineers	1.8	11,394
47-2231	Solar Photovoltaic Installers	1.8	5,719
29-9011	Occupational Health and Safety Specialists	1.6	47,336
11-2021	Marketing Managers	1.6	197,468
15-1133	Software Developers, Systems Software	1.5	218,581

B Data collection

Return-to-Office Data from Flex Index

I obtain information on firms’ return-to-office (RTO) policies by using a webscraper to collect the crowdsourced public information from the website Flex Index (<https://www.flexindex.com>).¹⁵

In total, I am able to collect the RTO data of 13,251 companies from the Flex Index site. This data includes each firm’s basic characteristics, including headquarter state and company size, and a classification of its office attendance requirements for employees.

After removing duplicates, I match the Flex Index companies to Compustat data to identify the standard company code (GVKEY) associated with the Flex Index companies—which are listed only by name. The matching process proceeds in the following order:

1. Fuzzy match company names, as well as state and/or city between the Compustat HQ location and the Flex Index company location.
2. Include all exact 1:1 company name matches in the final data
3. Drop imprecise matches that do not have city and state information
4. For remaining imprecise matches, drop all with fewer than 1000 employees, as such small firms are unlikely to be included in Compustat.
5. Manually review all remaining imprecise matches to see if both company name and location are plausibly the same, i.e. the company name only has generic differences, e.g. omitting “Corp.” or “Inc.”, and the company location is in the same or a neighboring metro area in both data sets. Include in the final data if both location and name are plausible matches.

The final dataset includes data on 1,336 companies with office requirements (8 Flex Index classifications, see Table 1) for which it was possible to determine a Compustat GVKEY, and Compustat company name.

See Table B.2 for details on the different RTO policy classifications. I classify a company as having provided a “Return-to-Office” mandate, if its RTO policy requires either a full-time presence, or requires a non-zero minimum number of days or time in the office (categories 3-8).

¹⁵The data was collected on March 3rd, 2025 by my research assistant Yi Li, who provided invaluable assistance in this task.

No.	Flex Index Category	Description
1	Fully Remote	Organization does not have offices; all employees work remotely.
2	Employee's Choice	Each employee can choose how often (or never) to come to the office.
3	Minimum Days a week	There is a minimum number of days employees must come to the office each week.
4	Specific Days a week	There are specific days of the week employees must come to the office.
5	Minimum & Specific Days a week	
6	Minimum % of Time per week	There is a minimum % of time that employees must be in the office.
7	Full Time in Office	Employees are expected to be in the office full time.
8	No Requirements	

Table B.2: Flex Index Return-to-Office Policy Classifications. Source: <https://www.flexindex.com/about>

C Additional results

A Determinants of remote work adoption

What type of firm is more likely to adopt remote work? The conceptual framework suggests that firms with greater technology skills should be more likely to invest in remote work technology. I test this prediction and the effect of other firm characteristics often used in the finance literature in regressions of the form

$$\text{RemoteWorkShare('21-'22)}_i = \alpha_{ind} + \beta \text{ FirmCharacteristic(2019)}_i + \text{Controls}_i + \varepsilon_i,$$

where all independent variables are standardized and the controls include the 2019 value of the dependent variable, so the coefficients can be interpreted as the effect of a standardized difference in the characteristic on changes in the remote work share. To proxy for the effect of simply hiring for roles that are more suitable for remote work, the control variables also include firm-level teleworkability in 2019 and 2021/2022, as well as the company's share of jobs requiring a college education in 2019, and NAICS 2-digit sector fixed effects.

The results are shown in Figure A.8: The first row shows the effect of a higher share of firm hiring in computer occupations on remote work adoption: firms that hire more for computer-related positions are more likely to go remote. The coefficients in rows 2-5 capture various dimensions of technology skill prevalence in pre-pandemic hiring of the firm and

show that hiring for data management, deep learning, machine learning, and data science skills predicts greater remote work adoption. While these findings are not causal, they are consistent with the predictions regarding a technological capability mechanism for greater remote work adoption in the model.

Other organizational and financial characteristics can provide context for understanding why some firms adopt remote work: the figure also shows that firms that are larger or less R&D-intensive, or that have a higher labor share or labor intensity in production¹⁶ are more likely to adopt remote work. With regard to financials, measures of higher earnings, such as gross profitability, ROA, and ROE, all predict higher remote work. However, as these measures require Compustat data, the sample size for these estimations is much smaller than for the main job posting sample and these results should therefore be considered only suggestive.

B Heterogeneity in generative AI Adoption by hiring characteristics

To close the loop on whether the characteristics that are engendered by greater remote work adoption then make it more likely that a firm in fact adopts generative AI, I split my sample based on the characteristics of a firm’s hiring in 2022 to see if particular hiring patterns lead to a stronger adoption of generative AI when a firm (or an occupation within a firm) has generative AI exposure.

I estimate cross-sectional specifications of the form

$$100 \times \text{GenAIJobShare}('23-'34)_i = \beta \text{GenAIExp}('21-'22)_i + \gamma \text{GenAIExp}('21-'22)_i \times \mathbb{1}[\text{High}_i] \\ FE_s + \text{Controls}_i + \varepsilon_i,$$

where the unit of observation i is either a firm, or an occupation-by-firm. Here, $\mathbb{1}[\text{High}_i]$ indicates whether a firm or an occupation-by-firm cell is above average in the characteristic in the sample. Some occupation characteristics are based on time-invariant 2018 O*Net data, which is the case for repetitiveness, social skills, interactiveness, inflexibility, decision-making intensity, decision-making without supervision, computer occupation, manager occupation, and leadership skills. The remaining occupation-by-firm level characteristics are means of hiring characteristics in 2022 Lightcast data. Firm-level and occupation-by-firm level hiring characteristics are computed as job posting-weighted means over job postings in 2022.

The results at the firm level are shown in Figure A.9, where the top panel shows esti-

¹⁶The latter is defined as $\ln(\text{Employment}/\text{PP\&E})$.

mates of $\hat{\gamma}$ that correspond to the difference in effects between the high and the low group, and the bottom panel shows the baseline effect $\hat{\beta}$ for the low group.

I find that firms that demand higher skills as proxied by education levels in their hiring and hire for occupations with less repetitive tasks are more likely to adopt generative AI if they have productivity potential from the technology. The same is true of firms that hired for more teamwork skills or for occupations that require social skills before ChatGPT was released. Firms that hire for jobs that normally require problem-solving and decision-making, or un-supervised decision-making, or for which the firm requires decision-making skills, are significantly more likely to adopt generative AI as well. I find the most pronounced difference in adoption between firms that have low and high technical skills such as machine learning or deep learning, with firms that hire more for these skills substantially more likely to adopt generative AI. Similarly, firms that hired more for managerial positions in 2022 were more likely to adopt generative AI.

Note that the types of positions that are associated with greater generative AI adoption are exactly the types of hiring that become more prevalent at the firm level with remote work adoption as shown in Figure 5. Together with the evidence in Figure A.9, this provides one potential mechanism for the technology ladder effect, which is that firms change their organizational structure and skill mix in response to remote work adoption, and this skill mix then proves useful for generative AI adoption. I discuss this connection in more detail in Section VI.

The corresponding results for within-firm differences in generative AI adoption in response to generative AI exposure at the occupation-by-firm level are shown in Figure A.10. The upper panel of the figure shows that, controlling for firm and occupation fixed effects, I find very similar patterns with regard to *which* occupations within the firm are more likely to adopt in response to exposure to generative AI: the only differences to the across-firm associations are that, within firms, the jobs requiring greater social skills or leadership skills are less likely to respond disproportionately to exposure, which aligns with the idea that these are positions that enable functioning communication and governance of technology-adopting firms, but in support of other roles that are more likely to be the ones directly working with the technology. Interestingly, the within-firm results also show that computer occupations are not more responsive to generative AI exposure, which suggests that the outsized adoption associated with technical skills is driven by other technical positions within the firm, e.g. more research-oriented analytical roles.

D Derivations

A Decision to go remote

Remote work is assumed to provide an additional productivity boost r_j to all the tasks done in an occupation, as shown in equation 3. This productivity boost can vary across occupations, such that there might be “teleworkable” jobs with a high r_j and other jobs where r_j is small or even negative (Dingel and Neiman, 2020). At the same time, remote work reduces the quality of decision-making support from management through triggering the communication barrier $\rho_f^{\mathbb{I}[\text{Remote}]} < 1$.

Based on the model above, a firm will adopt remote work (indexed by R) rather than stay non-remote (NR) if $\ln y_{fj}^R - \ln y_{fj}^{NR} > c_f^R$, which can be written as

$$\underbrace{(j - k_{fj})r_j}_{\text{Remote productivity bonus}} + \underbrace{\bar{\delta}_{[k,f]} \ln \left(\frac{Q_f^R}{Q_f^{NR}} \right)}_{\text{Output loss from } \Delta \text{ Decision-making quality}} > c_f^R$$

where R and NR again indicate optimal decisions conditional on going remote or not going remote. Here, $\bar{\delta}_{[k,f]}$ is the average decision intensity of human tasks at firm f with an automation level k and is defined as

$$\bar{\delta}_{[k,f]} = \frac{\int_k^j \delta(x) dx}{j - k} = \frac{e^{\lambda_s j} - e^{\lambda_s k_f}}{\lambda_s (j - k_f)}$$

Note that $Q_f^R < Q_f^{NR}$, such that the last term on the left is negative—remote work decreases overall decision quality, which represents a cost to be weighed against any potential productivity benefits.

B Remote work effect on generative AI adoption.

How does the adoption of remote work affect the decision of whether to adopt generative AI technology?

The difference in the net benefits from generative AI adoption between remote and non-

remote firms in this model is

$$\left[\ln y_{fj}^G - \ln y_{fj}^{NG} - c_f^G \right] \Big|_R - \left[\ln y_{fj}^G - \ln y_{fj}^{NG} - c_f^G \right] \Big|_{NR} \quad (D1)$$

$$= \underbrace{-(k^G - k_{fj})r_j}_{\text{Loss of remote productivity bonus (-)}} \underbrace{-(\bar{\delta}_{[k^G, f]} - \bar{\delta}_{[k, f]}) \ln \left(\frac{Q_f^R}{Q_f^{NR}} \right)}_{\Psi_f: \text{Relative human productivity effect (+)}} + \underbrace{(c_f^G \Big|_{NR} - c_f^G \Big|_R)}_{\text{IT complementarity (+)}}. \quad (D2)$$

As $(k^G - k_{fj})r_j > 0$, the first term implies that remote firms are disincentivized from adopting generative AI if they perceive large benefits from work-from-home, which have been assumed to scale with the share of tasks done by humans. The second term—the human output loss Ψ_f avoided due to generative AI adoption—is positive.

That is, as remote workers produce output with lower decision quality, there is less of a loss if generative AI produces the output in their stead. The last term reflects the fact that the cost of investing in generative AI c_f^{Gf} will be lower (and thus the adoption of generative AI more beneficial) if the firm previously adopted remote work, as I assume that remote work is the most recent technology wave before generative AI and therefore firms that adopt remote work have a higher level of information technology capital than those which do not.

Intuitively, this equation captures the trade-off that the work quality of remote workers is lower, so replacing it with generative AI is more profitable, and remote firms also already have some of the necessary infrastructure in place—but this has to be weighed against remote workers' longer effective work hours.

I make the assumption that the work-from-home productivity boost for the automated tasks is not too large relative to the technology investment cost savings for generative AI adoption when already having implemented remote work and the output quality differences among remote and non-remote workers, i.e. that

$$c_f^G \Big|_{NR} - c_f^G \Big|_R + \Psi_f > (k^G - k_{fj})r_j. \quad (D3)$$

If this holds, then we directly get the **the technology ladder effect**: *Firms that adopt remote work have a higher incentive to also adopt generative AI.*

If there is heterogeneity across firms, equation D2 also implies additional predictions with regard to when equation D3 is more likely to hold. The difference in output quality between non-remote and remote workers is larger when ρ_f is smaller, i.e. centralized decision support declines more with remote work. This means that firm organizational characteristics that can overcome these communication barriers (increase ρ_f) lower the quality loss when going remote, which then reduces the relative benefits of generative AI adoption

between remote and non-remote firms.

C Technology substitution: the effect of generative AI adoption on remote work

Does generative AI adoption impact a firm's incentive to continue using remote workers? The change in output from allowing remote work can be written as

$$\Delta_R(k_f) \equiv \ln y^R(k_f) - \ln y_{fj}^{NR}(k_f) = (j - k_f)r_j + (j - k_f) \underbrace{\bar{\delta}_{[k,f]}}_{\text{Avg. decision intensity}} \cdot \ln \left(\frac{Q_f^R}{Q_f^{NR}} \right). \quad (\text{D4})$$

We can then differentiate this with regard to k to obtain the effect of automation on this remote work benefit:

$$\frac{\partial \Delta_R}{\partial k} = -r_j + \underbrace{\frac{\partial \bar{\delta}_{[k,f]}}{\partial k}}_{\text{Automation effect on human judgment skew}} \cdot \ln \left(\frac{Q_f^R}{Q_f^{NR}} \right)$$

Here, $\ln \left(\frac{Q_f^R}{Q_f^{NR}} \right) < 0$, and the sign of $\frac{\partial \bar{\delta}_{[k,f]}}{\partial k}$ will be the same as the sign of λ_s . This means that the benefit of remote work is strictly decreasing in k_f for firms where automation has a low judgment skew, i.e. when $\lambda_s > 0$, and will be ambiguous if $\lambda_s < 0$. Unless automation with generative AI has a strong high judgment skew, i.e. substantially lowers the average decision intensity of human tasks, the average effects of automation on remote work should be negative.

To summarize, the model predicts a **technology substitution effect**: *Higher adoption of generative AI leads to a decline in the use of remote work technology for firms where automation has a low judgment skew, and has an ambiguous effect for firms where automation has a high judgment skew.* Note that, here I am assuming that for the purpose of the decision of whether to continue letting workers work remotely, the technical infrastructure is not relevant, as there is no way to “recoup” the original outlay for information technology capital to enable work-from-home when later eliminating remote work.

The size of this effect of generative AI adoption on remote work is scaled by the decision quality loss due to remote work adoption. This means that the same factors that reduce the impact of remote work on decision quality will also reduce the technology substitution effect:

- *Adoption of generative AI leads to a smaller decline in the use of remote work for firms*

with greater communication skills or managerial skills.

Moreover, note that if the level of decision intensity in different roles correlates with the judgment skew of automation, such that generative AI is more likely to automate relatively low judgment tasks if a role has a level of decision intensity, then we would also expect a greater technology substitution effect (negative effect on remote work) at firms with greater average decision-making intensity.

E Methodology notes

A Synthetic Difference-in-Differences

To estimate the effect of generative AI adoption on remote work, I need a way to create appropriate control and treatment groups that are matched on the observable remote work trends. An appropriate methodology in this case is therefore the synthetic difference-in-differences (SDID) design proposed by Arkhangelsky et al. (2021): instead of ex ante selecting which control variables might align remote work trends in the pre-event period between firms, this methodology first estimates which untreated firms have remote work trends that are most similar to the pre-trend of the treated units, and uses a weighted average of the most-comparable firms to create a “synthetic” control group. The estimation of post-event relative trends between the treated and control groups then proceeds analogous to a difference-in-differences estimation, but weighting potential control group firms based on the optimal weights, and also applying time weights that optimally put higher weights on pre-periods that are more comparable to post-event periods for the control group.

Conceptually, this corresponds to determining the treatment effect τ of being in a high generative AI exposure group in a regression equation of the form

$$\text{RemoteWorkShare}_{it} = \mu + \alpha_i + \gamma_t + \tau \mathbb{I}[\text{Post-ChatGPT}]_t \times \mathbb{I}[\text{High GenAI Exposure}]_i + \varepsilon_{it}, \quad (\text{E1})$$

which includes firm fixed effects α_i and time fixed effects γ_t , which absorb level differences between firms and national trends in the remote work share. The treatment effect τ is estimated with SDID by solving

$$(\hat{\tau}, \hat{\mu}, \hat{\alpha}, \hat{\gamma}) = \underset{\tau, \mu, \alpha, \gamma}{\text{argmin}} \left\{ \sum_{i=1}^N \sum_{t=1}^T (\text{RemoteWorkShare}_{it} - \mu - \alpha_i - \gamma_t - \tau \text{Treated}_{it})^2 \hat{\omega}_i^{\text{sdid}} \hat{\lambda}_t^{\text{sdid}} \right\} \quad (\text{E2})$$

where Treated_{it} stands for the interaction $\mathbb{1}[\text{Post-ChatGPT}]_t \times \mathbb{1}[\text{High Gen. AI Exposure}]_i$ between the post-release period indicator and the high generative AI exposure group assignment. The difference to conventional difference-in-differences comes from the optimal weights $\hat{\omega}_i^{\text{sdid}}$ and $\hat{\lambda}_t^{\text{sdid}}$ that are estimated in a first step to create a control group that matches the treatment group's trend in the remote work share (as opposed to differences in levels which are absorbed into the firm-level fixed effects).¹⁷

¹⁷I implement this estimation procedure using the `sdid` package in Stata, which is described in Clarke et al. (2023).